



LiDAR, IMU, and camera fusion for simultaneous localization and mapping: a systematic review

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Abstract

Simultaneous Localization and Mapping (SLAM) is a crucial technology for intelligent unnamed systems to estimate their motion and reconstruct unknown environments. However, the SLAM systems with merely one sensor have poor robustness and stability due to the defects in the sensor itself. Recent studies have demonstrated that SLAM systems with multiple sensors, mainly consisting of LiDAR, camera, and IMU, achieve better performance due to the mutual compensation of different sensors. This paper investigates recent progress on multi-sensor fusion SLAM. The review includes a systematic analysis of the advantages and disadvantages of different sensors and the imperative of multi-sensor solutions. It categorizes multi-sensor fusion SLAM systems into four main types by the fused sensors: LiDAR-IMU SLAM, Visual-IMU SLAM, LiDAR-Visual SLAM, and LiDAR-IMU-Visual SLAM, with detailed analysis and discussions of their pipelines and principles. Meanwhile, the paper surveys commonly used datasets and introduces evaluation metrics. Finally, it concludes with a summary of the existing challenges and future opportunities for multi-sensor fusion SLAM.

Keywords Multi-sensor · LiDAR · Camera · IMU · SLAM

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1 Introduction

Simultaneous localization and mapping (SLAM), which aims to obtain precise locations and reconstruct the unfamiliar environment, is a foundational technique for robots to perform high-level tasks, ranging from autonomous scenario exploration to emergency rescue operations. Intuitively, SLAM is a state estimation problem a state estimation problem that focuses on estimating the robot's trajectory and the map points in the environment. While odometry performs a similar task, its primary focus is on trajectory estimation. LiDAR-based odometry uses estimated trajectories to transform point clouds for map reconstruction. Unlike odometry, SLAM incorporates loop closure, trajectory optimization, and map refinement, making it a more comprehensive system. The most commonly used sensors in SLAM systems are Light Detection and Ranging (LiDAR) and camera, corresponding to two mainstream SLAM systems, i.e., LiDAR-based SLAM (L-SLAM) (Koide et al. 2019) and Vision-based SLAM (V-SLAM) (Mur-Artal et al. 2015).

L-SLAM relies on the intuitive ranging and point clouds acquisition capabilities of LiDAR to determine poses. It mainly reconstructs the surroundings by stacking point clouds based on the estimated trajectories. According to scan matching methods, LSLAM can be classified as direct and feature-based ones. Direct LSLAM (Frosi and Matteucci 2022; Koide et al. 2019) achieves scan matching by full point clouds, which mainly through Iterative Closest Point (ICP) (Umeyama 1991) and Normal Distributions Transform (NDT) (Biber and Strasser 2003). Feature-based LSLAM first extracts features of the two sets of point clouds and then matches the corresponding features (Shan and Englot 2018). Although L-SLAM has been widely used in academia and industry, it contains certain restrictions. For instance, LiDAR measurements lack color information and are not intuitive enough to extract semantic information. Besides, multi-line LiDARs with dense measurements are usually costly. In addition, L-SLAM is prone to fail in environments with degenerate geometries (eg. long tunnels, wide open spaces) and high dynamics (eg. moving obstacles and aggressive self-motion) (Tagliabue et al. 2021).

V-SLAM, which conducts SLAM based on visual sensors, estimates poses by multi-view geometry and recovers the environment by either simply stacking or implementing a full structure-from-motion (SFM) (Zhu et al. 2022b). Since cameras can provide ample data, such as dense color information and fine-grained texture, researchers have developed many V-SLAM approaches. According to the type of information employed in the system, VSLAM methods can be classified into direct (Engel et al. 2018), feature-based (Mur-Artal et al. 2015), and semi-direct VSLAM (Forster et al. 2014) methods. However, direct and semi-direct methods both require accurate photometric calibrations and the direct methods are sensitive to rapid sunlight changes (Yang et al. 2018). Moreover, monocular cameras are ambiguous in depth sensing, which hinders initialization and scale calculations. Despite the depth perception ability of stereo and RGB-D cameras, the distance they can detect is relatively limited. In addition, visual sensors struggle in feature extraction with low-lighting, texture-less and high-occlusion conditions. These flaws limit the adoption and accuracy of V-SLAM (Fuentes-Pacheco et al. 2015).

On the other hand, the Inertial Measurement Unit (IMU) collects motion data directly from robots via accelerometers and gyroscopes at high frequency. In contrast to exteroceptive sensors, it is generally not affected by the environmental changes mentioned above. It can offer an accurate pose transformation in a short period of time, but suffers from long-

term error drifts, which require additional corrections from other sensors (Yang and Huang 2018). In addition, the magnetometer in IMU is susceptible to electromagnetic interference in the environment. Although the accelerometer and gyroscope measurements remain unaffected, the magnetometer may produce an erroneous Yaw angle, resulting in inaccurate overall output. In conclusion, the degeneration condition of the three sensors is in Table 1.

Obviously, these sensors can compensate for each other to some extent. LiDAR and IMU can help cameras overcome scale ambiguity and address low-light conditions. IMU can support the measurement if LiDAR or cameras fail due to challenging environments, with its integral drift corrected by LiDAR and cameras. In addition, cameras can enrich the environmental information from LiDAR measurements. Moreover, while advancements like Bird's Eye View (BEV) in autonomous driving highlight progress in multi-sensor integration, fully leveraging the complementary properties of diverse sensors remains a challenge. This emphasizes the necessity of research into multi-sensor fusion SLAM methods.

Current multi-sensor fusion SLAM systems are primarily based on the above-described sensors. Thus, we mainly focus on systems supported by LiDAR, camera, and IMU. They can be categorized by fused sensors into four types: LiDAR-IMU-based SLAM (LI-SLAM) (Geneva et al. 2018; Ye et al. 2019; Palieri et al. 2021; Li et al. 2021b; Xu et al. 2022a), Visual-IMU-based SLAM (VI-SLAM) (Qin et al. 2018; Huai and Huang 2018; Yang and Huang 2019; Campos et al. 2020; Yu et al. 2021; Stumberg and Cremers 2022), LiDAR-Visual-based SLAM (LV-SLAM) (Huang et al. 2020; Zhang et al. 2017; Graeter et al. 2018; Shin et al. 2018; Young et al. 2020), and LiDAR-IMU-Visual-based SLAM (LIV-SLAM) (Zuo et al. 2019, 2020; Jia et al. 2021; Zhao et al. 2021; Lin and Zhang 2022; Zheng et al. 2022; Lang et al. 2023).

Although multi-sensor fusion SLAM inherits the advantages of single-sensor SLAM and can facilitate more diverse application scenarios, its practical implementation is pretty complex due to the heterogeneity of multiple sensors. To overcome this, several critical issues need to be paid attention to, including fusion methods, data preprocessing, and calibration. In addition, it also encounters challenges that disturb prevalent SLAM systems, such as dynamic and degenerative environments. The four types of multi-sensor fusion SLAM systems have similarities and differences in handling these challenges, while each of them has its problem to solve. Therefore, it is essential to recognize the exclusive issues in multi-sensor fusion SLAM and its exceptional treatments for common SLAM difficulties. We systematically summarize the essential basics of the multi-sensor fusion SLAM, as well as analyze the advantages and disadvantages of systems with different sensor fused types. Furthermore, we investigate the implementation of deep learning in multi-sensor fusion SLAM, which has not been fully explored in this field. Finally, we discuss the potential direction of the future development and improvement of multi-sensor fusion SLAM.

Several relevant works survey multi-sensor fusion SLAM. Most of them only focus on one type of sensor fusion, such as camera and IMU (Huang 2019), camera and LiDAR

Table 1 The degradation scenarios of LiDAR, IMU, and Camera

Sensor	Degeneration conditions
LiDAR	plane-only Structure; long corridors; vast open fields; reflective surfaces; obstacles on the ground
IMU	slow or stable movements; electromagnetic interference environment
Camera	aggressive motions; rapidly changing light conditions; texture-less environments; drastic motion

(Chghaf et al. 2022), and 3D LiDAR-based combinations (Xu et al. 2022b), thus lacking a systematic analysis of the varied compositions among these three sensors. Ebadi et al. (2024) survey the extraordinary works in *DARPA Subterranean Challenge*,¹ which is mainly focused on engineering practices of multi-sensor fusion SLAM for this specific competition. While some works contain multi-sensor fusion (Zhu et al. 2024a), they merely concentrate on conventional algorithms and lack investigation of learning-based methodologies. We additionally survey learning-based multi-sensor fusion SLAM and group it as a fusion framework, demonstrating new possibilities for end-to-end approaches. This paper aims to provide a systematic and comprehensive overview of the multi-sensor fusion SLAM systems. We conducted a survey of multi-source fusion methods based on LiDAR, IMU, and cameras in the field of robotics over the past six years, as these three sensors are currently among the most commonly used. It contains the infrastructural issues and analysis of four fused types of multi-sensor fusion SLAM, as well as their connections and differences.

The remaining of this paper is organized as follows: Sect. 2 introduces preliminary knowledges of multi-source fusion SLAM, including the framework, sensor coupling, fusion strategies, Lie algebra, sensor residual models, map representations, and calibration. Four categories of multi-sensor fusion SLAM algorithms are reviewed and analyzed in detail in Sect. 3, focusing on key methods and research challenges. The popular datasets and evaluation metrics are involved in Sect. 4. Finally, challenges and promising research directions are discussed in Sect. 5, and Sect. 6 concludes the paper. The whole framework of this paper can be shown as Fig. 1.

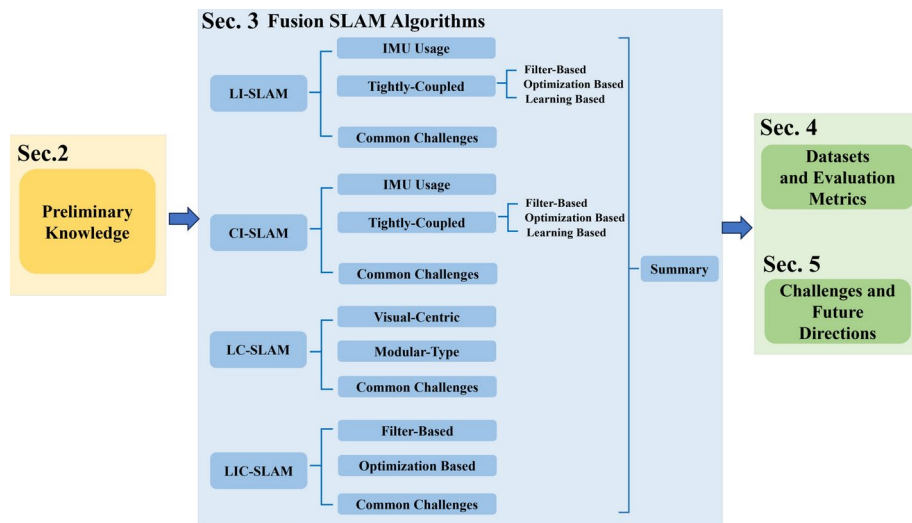


Fig. 1 The overview of this review. Sect. 2 introduces the basics of multi-source fusion SLAM, including its framework, sensor coupling methods, fusion techniques, Lie algebra, sensor residual models, map representations, and calibration. Sect. 3 categorizes and analyzes multi-source fusion SLAM algorithms, highlighting representative methods and key research issues. Sect. 4 reviews common datasets and evaluation methods, while Sect. 5 discusses future research directions and challenges

¹ <https://www.darpa.mil/program/darpa-subterranean-challenge>.

2 Preliminary knowledge

This section provides the basic background of multi-sensor fusion SLAM, disassembles the SLAM framework for the multi-sensor case, and introduces the implementation requirements.

2.1 SLAM framework

Multi-sensor fusion SLAM can be separated into *front-end* and *back-end*. The front-end extracts valid data and employs pre-processing, while the back-end infers and optimizes based on the data output by front-end. In Fig 2, we conclude the basic framework of multi-sensor fusion SLAM, which will be further elaborated in the following parts.

2.1.1 Front-end

The front-end is composed of data preprocessing, system initialization, data association, and loop closure. Data preprocessing is retrieving beneficial measurements from diverse heterogeneous data, including calibration, undistortion, and feature extraction. System initialization refers to obtaining the initial state of the system, such as scales and IMU bias. Data association addresses the spatial correspondence of the camera and LiDAR observations. Loop closure (LC) detects correlations in fresh and historical measurements and establishes constraints based on them to correct for drift.

2.1.2 Back-end

The back-end integrates and utilizes information across the board to acquire global consistent pose estimation and reconstructs the map of the environment. The state vector X to be

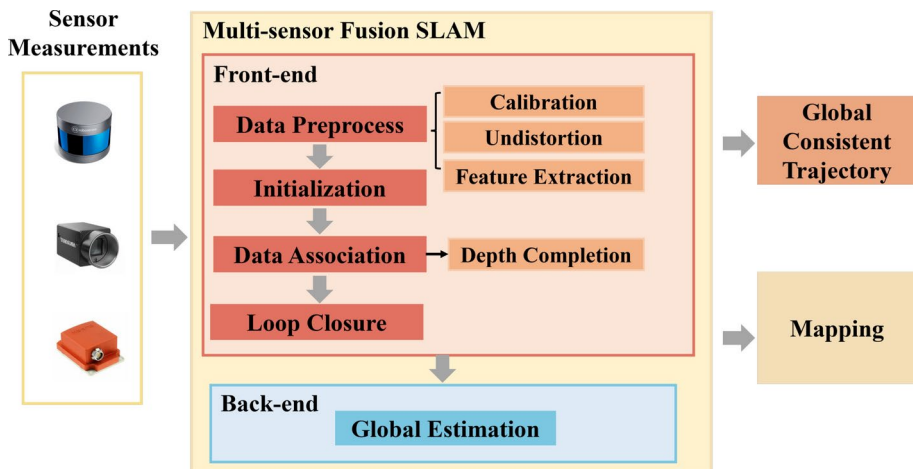


Fig. 2 The framework of multi-sensor fusion SLAM. The content within the red box represents the front-end, which includes data preprocessing, system initialization, data association, and loop closure detection. The content within the blue box represents the back-end, where the back-end process performs a global estimation of the system's state, providing a globally consistent map and trajectory measurements

estimated can be shown in (1). It usually contains the position Gp and rotation GR of the sensing platform, with other estimable variables involved in a multi-sensor fusion SLAM system, such as the linear velocity Gv , gravity vector Gg , and accelerometer and gyroscope biases b from IMU, a set of map points GP , extrinsic matrix T_{calib} between sensors, and the inverse depth λ of m image points. The upper-left corner marker $\{G\}$ indicates that the variation is relative to the world coordinate system.

$$X = [{}^GR, {}^Gp, {}^Gv, {}^Gg, b, {}^GP, T_{calib}, \lambda_0, \dots, \lambda_m] \quad (1)$$

2.1.3 Loop closure

Since loop closure (LC) is a complex component of the front-end, we provide a brief extension to introduce it. LC corrects accumulated drift by recognizing previously visited locations, ensuring global consistency in trajectory and map estimation. It typically involves three steps: **detection, validation, and optimization** (Tsintotas et al. 2022). Detection identifies potential revisits by comparing current and past data. Validation eliminates false positives using geometric or statistical methods. Optimization refines the trajectory by incorporating validated closures into a pose graph. Common approaches include feature-based methods using descriptors and Bag-of-Words (Galvez-Lopez and Tardos 2012), geometric methods such as ICP (Dellenbach et al. 2022) and multiview-stereo, and learning-based methods that leverage deep networks (Chen et al. 2021b; Cattaneo et al. 2022; Arandjelovic et al. 2016) for improved robustness.

2.2 Coupling method

Multi-sensor fusion SLAM systems can be broadly classified as **loosely-coupled and tightly-coupled** based on when the measurements are fused.

2.2.1 Loosely-coupled

The loosely-coupled systems handle data from different sensors individually and focus more on combining the intermediate estimations instead of raw measurements. Fusion on the intermediate estimations decreases the computation load and makes the loosely-coupled manner resilient and manageable. Hence, the loosely-coupled systems are generally applied to withstand sensor degradation or noise interference (Palieri et al. 2021; Han et al. 2023; Shao et al. 2019; Qin et al. 2020b; Beauvisage et al. 2022; Khattak et al. 2020; Khedekar et al. 2022).

Although loosely-coupled sensor fusion is appealing for runtime and robustness, it ignores correlations among diverse modalities and may cause information loss, convergence issues, and inaccurate estimation. Moreover, in extreme environments, the intermediate estimations of some sensors will fail and cause unreliable fusion in later stages.

2.2.2 Tightly-coupled

The tightly-coupled approaches directly fuse measurements of sensors to avoid the decoupling of constraints. Benefiting from the capitalization on the full range of heterogeneous data, the tightly-coupled systems generally have high accuracy. They heavily depend on precise models of sensors and their noise (Ebadi et al. 2024). To achieve better performance, the majority of multi-sensor fusion SLAM systems favor the tightly-coupled approach (Zhao et al. 2019; Qin et al. 2020a; Xu and Zhang 2021; Bai et al. 2022; Campos et al. 2021; Qin et al. 2018; Huai and Huang 2018; Qin et al. 2019; Heo et al. 2019; Huang et al. 2020; Wisth et al. 2021; Jia et al. 2021; Shan et al. 2021; Lin and Zhang 2022).

2.3 Fusion frameworks

Coupling methods determine when the measurements are fused while fusion frameworks indicate how the messages are integrated, which can be categorized as *filter-based*, *optimization-based*, and *learning-based*. Both loosely-coupled and tightly-coupled systems can use either filter-based or optimization-based fusion frameworks, yet learning-based fusion mainly exists in tightly-coupled systems.

2.3.1 Filter-based

Filtering methods in SLAM are primarily divided into EKF and particle filter (Nie et al. 2020), with EKF dominating filter-based multi-sensor fusion SLAM systems. EKF consists of two steps: propagation by motion prediction and updating by observation equations. The pipeline of common filter-based LI/VI-SLAM can be concluded as Fig. 3. IMU measurements are always propagated forward for state estimation. Exteroceptive sensors' measurements are responsible for calculating the residuals to update states until the system converges. In addition, IMU measurements are generally adopted for motion distortion of LiDAR and rolling shutter cameras (RS cameras). It is important to note that global shutter cameras (GS cameras) are generally less affected by motion distortion compared to the RS ones, as they capture the entire scene at once rather than line by line. Therefore,

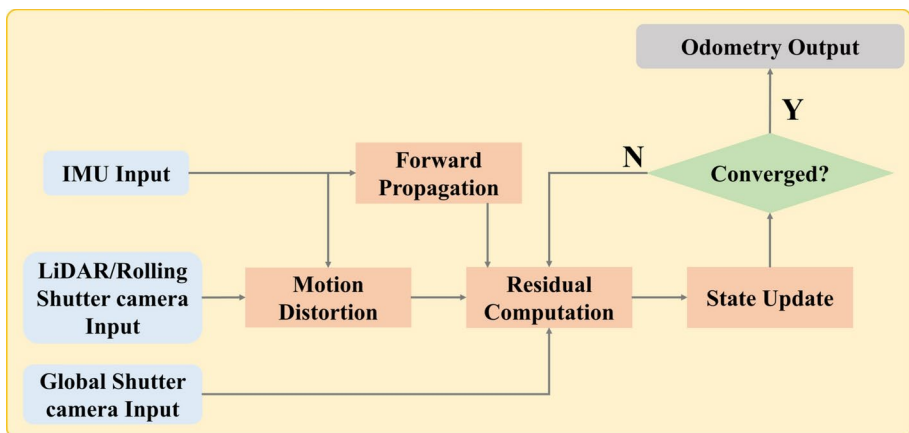


Fig. 3 The pipeline of filter-based LI/VI SLAM systems

most VSLAM methods require GS cameras for better performance. However, GS cameras typically involve higher cost and power consumption compared to RS cameras (Yang et al. 2018).

Filter-based systems are computational and memory efficient, thus widely applied in miniaturized embedded systems (Zhao et al. 2019; Qin et al. 2020a; Xu and Zhang 2021; Xu et al. 2022a; Bai et al. 2022; Huai and Huang 2018; Geneva et al. 2020; Qin et al. 2020b; Van and Mahony 2023; Wisth et al. 2021; Jia et al. 2021; Wang et al. 2019; Shan et al. 2021). However, there are several limitations. Since filter-based methods process measurements sequentially, they are sensitive to time synchronization and cannot correct past states directly using feature measurements. Current systems typically address this challenge by employing lag smoothers or re-propagating the state in the presence of delayed measurements, which helps to improve state estimation accuracy over time. In addition, traditional EKF generally suffers from inconsistency caused by linearization error and observability mismatch. Unscented Kalman Filter (UKF) (Wan and Van Der Merwe 2000) and Error State Kalman Filter (ESKF) (Sola 2017) mitigate this limitation to some extent. For nonlinear systems, UKF represents the state uncertainty by its approximate mean and standard deviation, which avoids linearization of kinematic and observation equations, as well as computation of Jacobians (Bahraini 2020). However, it still have some shortcomings, since the design of scaling parameters may cause approximation error. Some researchers propose Adaptive UKF that adaptively find suitable scaling parameters (Bozorg et al. 2019; Bahraini 2020). ESKF regards error states as state variables, offering minimal representation of state variables. Besides, it operates near the origin and avoid linearization approximation issues and gimbal lock problems (Wang et al. 2023a). These drawbacks of filter-based methods result in worse accuracy than optimization-based methods as revealed in (Strasdat et al. 2012).

2.3.2 Optimization-based

Different from filter-based approaches that incrementally estimate state, optimization-based methods jointly process numerous observations through nonlinear least squares optimization. This process is also called Bundle Adjustment (BA). The formulation can be expressed as follows:

$$\mathcal{X}^* = \arg \min_{\mathcal{X}} \left\{ \sum_{i \in \mathcal{I}} \|r_i\|_{\Sigma_I}^2 + \sum_{l \in \mathcal{L}} \|r_l\|_{\Sigma_L}^2 + \sum_{c \in \mathcal{C}} \|r_c\|_{\Sigma_C}^2 + \sum_{o \in \mathcal{O}} \|r_o\|_{\Sigma_O}^2 \right\} \quad (2)$$

where $\|x\|_M^2 = x^T M x$, $\mathcal{X}$ represents a set of state vectors. $r_{(\cdot)}$ and $\Sigma_{(\cdot)}$ are respectively the residuals and covariance of different measurements. The subscripts represent sensors they belong to. $\{i\}$, $\{l\}$, $\{c\}$ correspond to IMU, LiDAR, and camera respectively, and $\{o\}$ stands for other measurements, including other sensors, *priori*, and loop closure. The decorated letters denote a set of measurements. Equation 2 is to identify a set of state vectors that minimize individual residual terms, which will be interpreted in Sect. 2.5.

There are various solutions to optimization problems, such as graph optimization or the CasADi library (Andersson et al. 2019). Optimization-based multi-source fusion SLAM systems typically employ factor graph optimization, as it is one of the mainstream solutions for SLAM problems. In a factor graph, nodes represent variables to be optimized and edges indicate the constraints on the linked variables, which are formed by measurements. The

edges are also called factors. Take tightly-coupled optimization-based LI/VI SLAM systems for example, as shown in Fig. 4 the raw measurements of LiDAR and camera provide residuals constraints between map points and robot poses, while the IMU preintegration residuals constraint the transformation between consecutive frames. Since the flexibility of the factor-graph representation and the superior accuracy of BA, optimization-based methods have been broadly adopted in multi-sensor fusion SLAM systems (Geneva et al. 2018; Ye et al. 2019; Shan et al. 2020; Le Gentil et al. 2021; Nguyen et al. 2021; Li et al. 2022; Campos et al. 2021; Mascaro et al. 2018; Qin et al. 2018; Xie et al. 2020; Seok and Lim 2020).

However, as the trajectories continue to accumulate, the variables involved in the global BA will gradually become numerous, creating heavy computational burden and large storage requirements for the overall optimism. Prevalent methods alleviate this problem by tricks like sliding windows (Qin et al. 2018; Shan et al. 2020) and keyframes (Shan et al. 2020).

The sliding window limits batch optimization to a fixed number of frames within the window for bounded computational complexity. As measurements accumulate, new frames enter the window to participate in the optimization and old frames are marginalized. The marginalization approximates the informational content of the removed part of the graph as a prior on remaining nodes in the graph. This relationship will be used as a priori information to participate in the subsequent optimization of the window (Leutenegger et al. 2015). The highly optimized solvers specifically designed for BA (e.g., G2O (Kummerle et al. 2011), Ceres (Agarwal et al. 2022)) and incremental solvers (Kaess et al. 2008; Dellaert and Contributors 2022) are commonly adopted to accelerate the computation. In addition, pose graph is often used in the backend of optimization-based methods. Unlike BA that jointly optimize the robot poses and map points, the nodes of pose graph only represent for robot poses and the constraints are odometry measurements (Grisetti et al. 2010). Therefore, pose graph optimization is less computationally intensive compared to the full BA and is often used for global trajectory optimization (Qin et al. 2018).

Nonetheless, optimization-based methods will still require a bit more computation overall than filter-based methods, especially on devices such as small UAVs where computational resources are severely limited.

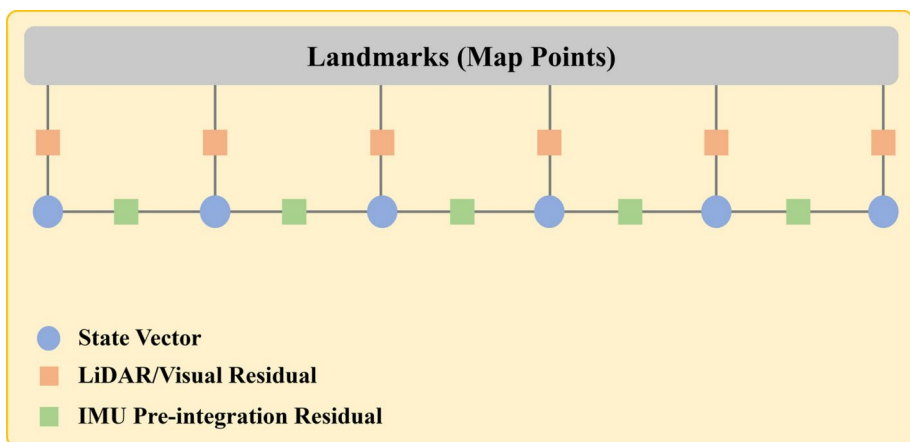


Fig. 4 The pipeline of tightly-coupled optimization-based LI/VI SLAM systems

2.3.3 Learning-based

The learning-based fusion multi-sensor SLAM methods we are concerned with refer to the fusion of features from multiple sources through neural networks, and then obtaining position and map results in an end-to-end data-driven way. However, the applications of multi-source end-to-end systems nowadays are generally limited to position estimation (Clark et al. 2017; Han et al. 2019; Shamwell et al. 2018a; Sun et al. 2023; Kurt et al. 2024). Therefore we only focus on the problem of multi-source fusion trajectory estimation in an end-to-end framework.

Learning-based multi-sensor fusion SLAM has three main problems to be solved, i.e., **feature extraction, feature fusion, and pose regression**. Feature extraction networks usually depend on the characteristics of the input data. For example, CNN is usually used to condense camera and LiDAR data, and LSTM is used to capture IMU information (Clark et al. 2017). Feature fusion is mostly achieved by LSTM (Clark et al. 2017; Chen et al. 2019). With the population of Transformer, more works have tried to utilize the attention mechanism in sensor fusion (Tu et al. 2022; Sun et al. 2023; Kurt et al. 2024; Liu et al. 2021). Pose Regression utilize the fused feature to infer position, which is typically implemented through fully connected layers (Tu et al. 2022; Clark et al. 2017; Chen et al. 2019).

While most learning-based methods have not yet demonstrated the same accuracy and generality as traditional methods, DNNs can extract implicit features from the input data. With the help of these implicit features, existing end-to-end methods (Tu et al. 2022) have been able to perform consistently in challenging environments where traditional methods (Qin et al. 2018) fail. In addition, end-to-end learning approaches (Clark et al. 2017; Chen et al. 2019) offer the potential to bypass challenges of rigorous initialization and calibration by directly learning fuse sensory information (Kurt et al. 2024). This highlights the potential for the application of end-to-end approaches and therefore this paper also contains an overview of learning-based fusion methods.

In summary, each fusion framework has its advantages and disadvantages, which lead to algorithmic trade-offs for different situations.

2.4 Lie group and lie algebra

Lie groups and Lie algebras (Chirikjian 2011) are fundamental in SLAM for efficiently representing and optimizing spatial transformations. Lie groups like $SO(3)$ and $SE(3)$ offer a compact, singularity-free framework for rigid-body motions, while their algebras, $\mathfrak{so}(3)$ and $\mathfrak{se}(3)$, enable linearization of transformations. Exponential and logarithmic mappings bridge nonlinear group operations and linear tangent space computations, enhancing optimization efficiency and stability. This approach ensures consistent handling of pose estimation, error propagation, and uncertainty modeling in SLAM systems. Here, we give a brief introduction to their definitions as well as transformations, more computational details can be found in (Chirikjian 2011) and Manif library.²

²<https://github.com/artvis/manif>.

2.5 Residual of different sensors

2.5.1 LiDAR

LiDAR is an active sensor, whose measurements are mainly composed of the intensities and 3D coordinates of point clouds in the LiDAR frame. The measurements of LiDAR are generally used as residuals from the scan-to-map or scan-to-scan matching. As mentioned in Sec. 1, L-SLAM can be categorized as direct and feature-based methods, resulting in different structure of residuals r_l .

Direct L-SLAM generally relies on scan registration methods such as ICP Umeyama (1991). It iteratively minimize geometric errors, such as point-to-point distances to achieve precise alignment. Recent direct LiDAR odometry KISS-ICP (Vizzo et al. 2023) also implements this simple yet effective point-to-point ICP and achieves considerable trajectory estimation. Thus, we take point-to-point distance as residuals r_{point} of Direct L-SLAM, which can be defined as:

$$r_{point} = \sum_i \|q_i - (Rp_i + t)\|^2 \quad (3)$$

where $\mathcal{P} = \{p_i\}$ and $\mathcal{Q} = \{q_i\}$ represent for two different sets of point clouds, respectively. (R, t) is the transformation between them.

Feature-based L-SLAM generally extracts line or plane features for transformation estimation. The line feature residual is the distance between observed points in the point cloud and the fitted line, while the plane feature residual is the distance between observed points and the fitted plane. Both residuals aim to optimize the system's pose by minimizing the distance between points and the line, or between points and the plane. Thus, the residuals are expressed as point-to-line distance r_{line} or point-to-plane distance r_{plane} .

$$\begin{aligned} r_{line} &= l \times ({}^G_L R^L P + {}^G t) + p_0 \\ r_{plane} &= n^T ({}^G_L R^L P + {}^G t) + d \end{aligned} \quad (4)$$

where ${}^G_L R$ and ${}^G t$ denote the rotation and translation of LiDAR; ${}^L P$ is a point cloud scanned in LiDAR frame; l is the direction vector of the line; p_0 is a point on the line; n^T and d denote the normal vector of the plane and the distance the plane to the origin, respectively.

According to the scan patterns, LiDARs can be clarified as mechanical-based and solid-state-based. The distinction of scan patterns results in different feature definition methods. Systems with mechanical-based LiDARs (Geneva et al. 2018; Ye et al. 2019; Zhao et al. 2019; Shan et al. 2020) commonly adopt the curvature calculation proposed in LOAM (Zhang and Singh 2014). However, it is not suitable for solid-state LiDARs. Take Livox MID40 LiDAR as an example, it has a small conical-shaped FOV in the front and adopts a non-repetitive rosette-like scanning pattern (Lin and Zhang 2020). Lin and Zhang (2020) traversed along the incident and deflection angle to choose point candidates for Livox MID40.

2.5.2 IMU

IMU can measure the sensor's linear acceleration a and angular velocity ω with bias b and noise n .

$$\begin{aligned} {}^B\tilde{a} &= {}^G R^T ({}^G a - g) + b_a + n_a \\ {}^B\tilde{\omega} &= {}^G \omega + b_\omega + n_\omega \end{aligned} \quad (5)$$

To simplify the presentation, we omit the upper-left corner label in the subsequent equations. Following the kinematic model (Eq. 6) in (Murray et al. 2017), the state, including rotation R , velocity v , and position p at t_j can be obtained by integrating Eq. 6, as shown in 7.

$$\begin{aligned} \dot{R} &= R\omega^\wedge, \quad \dot{v} = a, \quad \dot{p} = v \\ R_j &= R_i \text{Exp}\left(\int_{t_i}^{t_j} \omega(\tau) d\tau\right) \\ v_j &= v_i + \int_{t_i}^{t_j} a(\tau) d\tau \\ p_j &= p_i + \int_{t_i}^{t_j} v(\tau) d\tau + \iint_{t_i}^{t_j} a(\tau) d\tau^2 \end{aligned} \quad (6)$$

$$\quad (7)$$

Assuming a and ω remain constant during time interval (t_i, t_j) and combining the measurement model, we can obtain that:

$$\begin{aligned} R_j &= R_i \text{Exp}\left((\tilde{\omega}_i - b_i^\omega - \mathbf{n}_i^{\omega d}) \Delta t_{ij}\right) \\ v_j &= v_i + g\Delta t_{ij} + R_i (\tilde{a}_i - b_i^a - \mathbf{n}_i^{ad}) \Delta t_{ij} \\ p_j &= p_i + v_i \Delta t + \frac{1}{2} g \Delta t_{ij}^2 + \frac{1}{2} R_i (\tilde{a}_i - b_i^a - \mathbf{n}_i^{ad}) \Delta t_{ij}^2 \end{aligned} \quad (8)$$

where n_i^{ad} and $n_i^{\omega d}$ represent for discrete-time noises. However, every time the poses are adjusted, this propagation needs to be repeated, which is time consuming (Lupton and Sukkarieh 2012). The IMU *preintegration* model avoids repropagation by considering the integration between $[t_i, t_j]$ as virtual measurement, including preintegrated rotation mea-

surement $\Delta\tilde{R}_{ij}$, preintegrated velocity measurement $\Delta\tilde{v}_{ij}$, and preintegrated position measurement $\Delta\tilde{p}_{ij}$. This model is only related to IMU measurements instead of states in t_i and t_j (Forster et al. 2017), which can be shown as:

$$\begin{aligned} \Delta\tilde{R}_{ij} &= R_i^T R_j \text{Exp}(\delta\phi_{ij}) \\ \Delta\tilde{v}_{ij} &= R_i^T (v_j - v_i - g\Delta t_{ij}) + \delta v_{ij} \\ \Delta\tilde{p}_{ij} &= R_i^T \left(p_j - p_i - v_i \Delta t_{ij} - \frac{1}{2} g \Delta t_{ij}^2 \right) + \delta p_{ij} \end{aligned} \quad (9)$$

where $\delta\phi_{ij}$, δv_{ij} , and δp_{ij} represents for the noise of rotation, velocity, and position, respectively. Thus, the residual of IMU preintegration can be modeled as:

$$\begin{aligned} r_{\Delta R_{ij}} &= \text{Log} \left(\Delta \tilde{R}_{ij}^T R_i^T R_j \right) \\ r_{\Delta v_{ij}} &= R_i^T (v_j - v_i - g \Delta t_{ij}) - \Delta \tilde{v}_{ij} \\ r_{\Delta p_{ij}} &= R_i^T (p_j - p_i - v_i \Delta t - \frac{1}{2} g \Delta t_{ij}^2) - \Delta \tilde{p}_{ij} \end{aligned} \quad (10)$$

Although the classical IMU preintegration formulas described above have been widely adopted in many systems (Qin et al. 2018; Shan et al. 2020, 2021), it is still limited to Euler integration for position and velocity where the orientation is assumed non-rotating between IMU measurements (Wang et al. 2024). Researchers have done some works to improve the original IMU preintegration. For instance, MAVIS (Wang et al. 2024) extends the formulas based on the exponential function of an automorphism of $\text{SE}_2(3)$, which effectively improves the tracking performance during rapid motions and extended integration times. Hu et al. (2022) proposes a backward IMU preintegration method that obtains states from states at later moments. The switching between classical and backward preintegration helps the system to find more accurate correlations between asynchronous data.

2.5.3 Camera

Feature-based methods and direct methods are two fundamental paradigms of V-SLAM. Feature-based methods estimate motion by extracting and matching sparse image features, optimizing poses and structure through reprojection error minimization. While robust under significant inter-frame motion due to reliable feature descriptors, they are prone to drift and rely heavily on detection accuracy. Direct methods, on the other hand, optimize photometric error directly from image intensity values, leveraging all image information, including low-gradient areas, making them more robust in low-texture scenes. Although computationally intensive due to image warping, direct methods eliminate the need for feature detection and descriptor computation, offering an alternative residual formulation (Forster et al. 2014).

Thus, the residual of camera measurements r_c can be categorized as geometric error r_g and photometric error r_p , corresponding to feature-based and direct V-SLAM, respectively. The geometric error relies on the imaging models of cameras such as the pin-hole model and MEI mode (Mei and Rives 2007). Express the projection function in terms of π , the reprojection error between matched 3D points P_i^G in the global frame and keypoints p_i with $i \in \mathcal{M}$ the set of all matches can be modeled as:

$$r_g = \sum_{i \in \mathcal{M}} \|p_i - \pi(RP_i^G + t)\|_{\Sigma}^2 \quad (11)$$

where R and t denote the rotation and translation of the camera; Σ is the covariance matrix associated with the scale of the keypoint.

The photometric error is based on the photometric consistency assumption and thus is more sensitive to illumination change. Define $I_{(\cdot)}(p_{(\cdot)})$ as the intensity of point P_j^G in dif-

ferent frames, the photometric constraints between the host frame m and the target frame n can be derived as:

$$r_p = \sum_{j \in \mathcal{N}} \|I_m(p_{m,j}) - I_n(p_{n,j})\|^2 \quad (12)$$

where $\mathcal{N}$ is the set of pixels included in the calculation.

It is worth mentioning that there exist some distinctions in feature parameterization of r_g and r_p . Reprojection error typically uses 3D coordinates for feature representation, while photometric error often relies on inverse depth derived from the first observation, enabling efficient optimization in monocular settings. Additionally, photometric error is sensitive to exposure variations (e.g., brightness, gain, exposure time), necessitating robust exposure transfer estimation, as exemplified in LSD-SLAM (Engel et al. 2014).

2.6 Mapping

The Mapping component can be broadly categorized into two formats: landmark-based maps and volumetric-based maps. Landmark-based maps represent the environment using sparse and distinct features such as points, lines, or planes, often extracted from sensory data. These maps are computationally efficient and well-suited for real-time applications, but they are generally sparse and lack detailed environmental geometric information (Mur-Artal et al. 2015; Shan et al. 2020). In contrast, volumetric-based maps provide a dense representation of the environment, typically constructed using techniques like Voxel Grids or Truncated Signed Distance Functions (TSDF) to capture the full geometric structure. While these maps offer a more detailed reconstruction suitable for complex environments, they demand higher computational and memory resources. Recently, with the population of Neural Radiance Fields (Mildenhall et al. 2021) and 3D Gaussian Splatting (3DGS) (Kerbl et al. 2023), many SLAM methods have adopted these high-fidelity volumetric-based map as their mapping modules. For example, NICER-SLAM (Zhu et al. 2024b) jointly optimizes poses and NERF map in an end-to-end manner, while MONO-GS (Matsuki et al. 2024) adopts 3DGS as the only underlying scene representation. However, the frame rates of these methods are still inferior to those of landmark-based mapping approaches.

2.7 Calibration

Multi-sensor calibration is a crucial prerequisite for multi-sensor fusion SLAM, as there are spatial and temporal discrepancies between the heterogeneous sensor data. Since calibration is not the main focus of this paper, we will provide a brief explanation of the basic concepts of calibration and list some open-source tools. We recommend that readers refer to (Qiu et al. 2023; Eckenhoff et al. 2019a) for a more comprehensive and in-depth understanding of multi-sensor calibration methods and principles.

Multi-sensor calibration includes spatial and temporal calibration. Spatial calibration determines the spatial relationship between sensors mounted on the same hardware, i.e., obtaining the extrinsic parameters to align heterogeneous data in a common coordinate system. For IMU-based systems, alignment of trajectory estimates over time is a typical approach (Lv et al. 2020; Zhu et al. 2022a). For LiDAR-camera systems, extrinsic param-

eters are estimated by aligning common features captured by both sensors (Yuan et al. 2021; Koide et al. 2023). We provide some common open-source toolboxes for spatial calibration in Table 2.

Temporal calibration refers to identifying the time offset t_d between sensor data caused by transmission delays or data processing (Li and Mourikis 2014). Most systems address t_d through hardware (e.g., FPGA (Wenzel et al. 2021)) or network protocols (e.g., Network Time Protocol³). After unifying the clock source, measurement differences caused by frequency variations are generally handled by linear interpolation (Xu and Zhang 2021).

The discussion above focuses on offline calibration, conducted independently of system operation. Despite its maturity and available tools, it requires recalibration each time hardware changes. Thus, many multi-sensor SLAM systems contain the online calibration function and we will discuss this in detail in Sect. 3.

3 Fusion SLAM algorithm

We mainly focus on the multi-sensor fusion SLAM algorithms in recent 6 years and classify the multi-sensor fusion SLAM systems according to the fused sensors as LI-SLAM, VI-SLAM, LV-SLAM, and LIV-SLAM. For each type of system, we first choose the appropriate classification method from the coupling approach or the fusion frameworks and then analyze their solutions for common SLAM challenges and specific problems.

3.1 LiDAR-IMU SLAM

LI-SLAM systems originate from L-SLAM systems. One of the milestones of feature-based L-SLAM algorithms is LOAM (Zhang and Singh 2014), which defines local curvature to extract edge and planar features. Many feature-based LI-SLAM works employ its feature extraction manner (Qin et al. 2020a; Shan et al. 2020; Xu and Zhang 2021; Li et al. 2021b; Lv et al. 2021; Jiang and Shen 2022).

Table 2 Spatial calibration toolboxes for LiDAR, IMU, and camera

Sensors	Tool box	References link
LiDAR & IMU	<i>LiDAR_align</i>	https://github.com/ethz-asl/LiDAR_align
	<i>LiDAR_IMU_Calib</i>	https://github.com/APRIL-ZJU/LiDAR_IMU_calib
	<i>LiDAR_IMU_Init</i>	https://www.github.com/hku-mars/LiDAR_IMU_Init
Camera & IMU	<i>kalibr</i>	https://github.com/ethz-asl/kalibr
LiDAR & Camera	<i>livox_camera_calib</i>	https://github.com/hku-mars/livox_camera_calib
	<i>direct_visual_LiDAR_calibration</i>	https://github.com/koide3/direct_visual_LiDAR_calibration
LiDAR & Camera & IMU	<i>Autoware</i>	https://github.com/autowarefoundation/autoware
	<i>OpenCalib</i>	https://github.com/PJLab-ADG/SensorsCalibration

³<https://datatracker.ietf.org/doc/html/rfc5905>.

Since LiDAR continuously scans the environment when the system keeps moving, points in one scan are obtained at different timestamps, resulting in motion distortion. In LI-SLAM systems, a universal application of IMU is the correction of LiDAR point clouds (de-skew) in the data preprocess stage of the front-end (Zhao et al. 2019; Shan et al. 2020; Palieri et al. 2021; Xu and Zhang 2021; Nguyen et al. 2021; Li et al. 2021c; Liu et al. 2022; Junior et al. 2022).

The majority of methods assume that time is discrete and undistort LiDAR points by interpolations. IMU locates the corresponding point clouds and projects them into a unified LiDAR scan according to the timestamped poses measured by it, which also contributes to the dynamic object filter (Zhao et al. 2019; Shan et al. 2020). However, discrete-time-based approaches have some inherent limitations. Their interpolations are accompanied by errors. Besides, handling ample raw measurement directly is intractable. To avoid these problems, researchers apply the continuous time model (e.g. B-spline (Sommer et al. 2020)) to involve the undistortion in trajectory estimation (Le Gentil et al. 2021; Lv et al. 2021; Chen et al. 2023b). Chen et al. (2023b) design a two-step coarse-to-fine point cloud deskew method by first coarsely constructing the trajectory through IMU integration and then refining it based on continuous-time equations. It is worth mentioning that distinct from the above methods, He et al. (2023) fuse each LiDAR point at its true sampling time rather than a scan to avoid motion distortion with IMU.

Early LI-SLAM works are mainly loosely-coupled that either simply regard the transformations from IMU as priors or directly integrate odometry results (Zhang and Singh 2014; Zhen et al. 2017). Nowadays loosely-coupled LI-SLAM approaches are more delicate in utilizing the resilience of the loosely-coupled manner, which enables modality change in specific conditions (Palieri et al. 2021; Reinke et al. 2022; Han et al. 2023).

To comprehensively use information from all sensors, popular LI-SLAM systems are predominantly tightly-coupled and can be categorized as *filter-based*, *optimization-based*, and *learning-based* ones according to the fusion framework.

3.1.1 Filter-based

The main drawback of filter-based approaches is the cumulative error since EKF is susceptible to linearization errors (Huang and Dissanayake 2007). General improvements attach importance to modifying the EKF, such as IEKF and iterated error-state Kalman Filter (IESKF). LINS (Qin et al. 2020a) presents a robot-centric IESKF, which recursively corrects the estimated state by generating new features in each iteration. Xu and Zhang (2021) exploit IESKF to reduce linearization errors and propose a new complexity-reduced formula to calculate the Kalman gain. Fasterlio (Bai et al. 2022) applies the Anderson acceleration to the iteration of IESKF (AA-IESKF) (Gao et al. 2022).

In addition to fixing the cumulative error, many works also focus on enhancing the computational efficiency of the filter-based LI-SLAM systems. The computation speed can be accelerated by optimizing the data structure of point clouds (Xu et al. 2022a; Bai et al. 2022; Reinke et al. 2022; Jiang and Shen 2022; Ji et al. 2024). The classical k-d tree structure requires time to build, iterate, remove unwanted nodes, and reconstruct the tree, which influences the efficiency. FAST-LIO2 (Xu et al. 2022a) develops incremental kd-tree (ikd-tree) (Cai et al. 2021) to represent dense point clouds. This novel data structure enables dynamic updates of the tree and reduces the computation cost of dense registration of point clouds,

which makes FAST-LIO2 a particularly outstanding achievement in the LI-SLAM field. The hash-based voxel structure can further speed up the system. Bai et al. (2022) present the sparse incremental voxel (iVox) with the help of the k-d tree and Hilbert curve. Ji et al. (2024) further propose an iVox that only contains a centroid and covariance matrix. They utilize the easy addition and deletion characteristic of voxels and ease the computation burden. Some other dynamic data structures such as octree have been also adopted to manage the map (Reinke et al. 2022; Jiang and Shen 2022).

3.1.2 Optimization-based

In addition to the IMU preintegration factor and LiDAR factor, some optimization-based LI-SLAM systems include other factors to improve their performance. The widely used one is the loop detection factor which acts as a constraint when the system revisits a place (Shan et al. 2020; Liu et al. 2022). Researchers also design novel factors for specialized constraints, such as new features (Geneva et al. 2018), gravity (Wang et al. (2023c)), dynamic tracking (Lin et al. 2023), and matching cost (Koide et al. 2022). Koide et al. (2022) devise a voxelized Generalized Iterative Closest Point (GICP) matching cost factor, which employs more points in the calculation of registration error than the conventional line and plane point matching.

To overcome the time-consuming drawback of the optimization-based method, researchers leverage critical components of the measurements rather than using the whole data in optimization (Shan et al. 2020; Li et al. 2021b; Nguyen et al. 2021; Lv et al. 2021; Koide et al. 2022). Shan et al. (2020) present LIO-SAM, a fairly classic and versatile optimization-based LI-SLAM system that utilizes key-frame, sliding window, marginalization, and local map. It selects a keyframe when the pose change exceeds a threshold relative to the previous state. The sliding window is used to create a local point cloud map for scan-matching and the old LiDAR scans are marginalized as priors. LiLi-OM (Li et al. 2021b) adopts a hierarchical keyframe-based sliding window optimization to fuse IMU and LiDAR measurements.

3.1.3 Learning-based

In contrast to the extensively studied conventional frameworks, there are relatively few learning-based LI-SLAM. Most of them exploit CNN for LiDAR and LSTM for IMU feature extraction (Son et al. 2021; Iwaszczuk et al. 2021; Tu and Xie 2021). Recent methods leverage the globality and generalisability of the Transformer to fuse information of different modalities. TransFusionOdom (Sun et al. 2023) projects LiDAR and IMU measurements in the image plane, which is suitable for CNN to extract features. It fuses the homogeneous and heterogeneous information through soft mask attention and Transformer respectively, and achieves quite encouraging results on the KITTI dataset (Geiger et al. 2012).

3.1.4 Common challenges

In addition to improvements tailored specially to deal with the shortcomings of fusion frameworks, LI-SLAM contains several solutions for dynamic, degeneracy, and online calibration and is often improved by enhancing feature extraction.

Dynamic: High dynamics issue combats the static hypothetical of SLAM systems. The environmentally independent nature of IMU relieves the influence of dynamic objects to a certain extent. A more reliable method is to combine DNNs in dynamic filtering. LIOM (Zhao et al. 2019) introduces Fully Convolutional Neural Networks (FCNN) for dynamic object removal and obtains static background point clouds. Apart from learning-based approaches, dynamic points can be detected according to the visibility assumptions. For instance, RF-LIO (lio 2021) is based on the assumption that a closer point in a narrow FOV occludes the farther point, while a distant point in the map implies the corresponding closer point in the query scan is moving.

These methods always treat dynamic points as disturbances and reject them after detected.

Degeneracy: Most LI-SLAM systems pay more attention to the degeneracy in LiDAR measurements caused by the environment. Several methods inherit the idea in (Zhang et al. 2016) that inspect degeneracy by the Hessian of LiDAR measurement (Shan et al. 2020; Tagliabue et al. 2021; Han et al. 2023). LION (Tagliabue et al. 2021) analyzes the Hessian of the point-to-plane ICP function and chooses another odometry by HeRO (Santamaria Navarro et al. 2019) when degeneracy is too large. Since the system runs in long tunnels, it deems the translation part to be the most challenging to estimate. Thus, its observability metric relies on the ratio between the largest and smallest eigenvalues of the translation component. The frequency of measurement is also a criterion for degeneracy. LOCUS (Palieri et al. 2021) contains a simple health monitor module that deems the input as healthy when the rate is sufficient (>1 Hz). Otherwise, it will choose another measurement according to its predefined priority queue of modalities. Some methods assess the quality and number of the detected features (Junior et al. 2022) or devise uncertainty models for LiDAR measurements (Jiang and Shen 2022). Jiang and Shen (2022) devise a principled uncertainty model and a noise estimation method for LiDAR measurements to mitigate the uncertainty estimation problem.

Online Calibration Online calibration optimizes calibration variants along with state vectors. IN2LAAMA (Le Gentil et al. 2021) involves time offset in IMU preintegration and builds a time-shift residual through IMU bias. The extrinsic is utilized when extracting features and is included in the feature residuals. It integrates time offset and extrinsic into state variables and calibrates by jointly optimizing the time-shift factor, feature factor, and IMU factor.

Feature: As for LI-SLAM built upon filter-based L-SLAM methods, feature extraction in the front-end is one of the preliminaries for the state estimation. In terms of craft geometric descriptors for edges and planes, the gravity and angular information from IMU is extremely helpful (Geneva et al. 2018; Li et al. 2021c; Wang et al. 2023b). LIPS (Geneva et al. 2018) proposes a novel formulation using the closest point plane representation to overcome the singularity issue of Hessian and generates an anchor plane factor. Li et al. (2021c) identify the ground plane based on the gravity vector from IMU. They determine corner points by calculating the angle between the target point and two adjacent points, which later assists the aggregation of lines and surfels. As for solid-state LiDAR, Li et al. (2021b) tailor a novel feature extraction method for Livox Horizon performed in the time domain. Some approaches make extra use of the intensity measured by LiDAR. Li et al. (2022) combine intensity and geometry to find edge points that help to resist the degeneracy of solid-state LiDAR. Zhang et al. (2023d) introduce reflectivity texture information to build photometric residuals blended with geometric residuals by IEKF. Screening for key components in all

features can also provide beneficial effects. KFS-LIO (Li et al. 2021d) refers to the Dilution Of Precision concept (Kaplan and Hegarty 2017) in satellite position. It introduces a quantitative LiDAR feature evaluation module that counts the Jacobian of the state error of position for feature selection.

3.2 Visual-IMU SLAM

Obtaining scale information is conducive to monocular cameras with scale ambiguity. Since IMU can provide measurements of ego-motion like accelerations and attitude, the general applications of IMU in VI-SLAM lay in metric scale recovery, which is typically implemented during the initialization of the systems.

The purpose of the initialization of VI-SLAM is to retrieve essential parameters of the system, including metric scale, gravity direction, velocity, and biases of IMU. Some methods integrate scale and gravity direction into the optimization model and initialize immediately with an arbitrary scale (Von et al. 2018). More generally, with the advent of VIN-MONO (Qin et al. 2018), the initialization method it employs has become a classical paradigm (Qin and Shen 2017; Qin et al. 2018; Seok and Lim 2020; Campos et al. 2021; Zhang et al. 2023c). It first performs an up-to-scale visual-only SFM and then loosely aligns it with the IMU preintegration in a restricted sliding window (Qin and Shen 2017). ORB-SLAM3 (Campos et al. 2021) improves this approach by exploiting an inertial-only initialization method which considers IMU measurement uncertainty before aligning the IMU trajectory and V-SLAM trajectory. It can achieve sophisticated initialization in less than 4 s (Campos et al. 2020). However, IMU is not able to determine a correct initialization if the motion is carried out through constant velocity. Stumberg and Cremers (2022) deal with this problem by continuously estimating the scale and gravity direction after initialization. They present delayed marginalization and pose graph bundle adjustment (PGBA) to alleviate the computing effort and inject IMU factors into marginalized states for IMU initialization. The robust and long-term promised scale and gravity direction estimation make DM-VIO perform better than VINS-MONO. In particular, robot-centric formulation avoids the need to align the initial orientation with the global gravitational direction and allows the VI-SLAM to start at any pose (Huai and Huang 2018).

Stereo cameras are not limited by scale ambiguity since they can calculate depth directly. Stereo-SLAM algorithms (Mur-Artal and Tardos 2017) can use this depth and generally performance better than pure monocular-SLAM. However, stereo cameras take up more computational resources than monocular cameras, while typically being larger in size. These limitations can make binocular cameras more difficult to deploy on platforms with limited resources and size, such as small UAVs. Their depth is limited by the baseline and they will inevitably encounter the same challenges that monocular cameras do in dynamic and textureless environments (Yu et al. 2023). Stereo-inertial SLAM focuses on other perspectives rather than depth, such as reducing computing consumption (Sun et al. 2018), stereo-IMU calibration (Yin et al. 2023b), IMU-assisted feature extraction (Liu et al. 2023), unsupervised method (Han et al. 2019), and utilization of scene flow (Yin et al. 2023a; Wen et al. 2024a). Stereo cameras bring more computation burden because of the additional camera included. Sun et al. (2018) alleviate this by representing the stereo observation in $\mathbb{R}^4$ to avoid the need for stereo rectification.

Analogous to LI-SLAM systems, early VI-SLAM systems are primarily loosely-coupled (Forster et al. 2015), while the tightly-coupled ones are dominantly researched today. We analyze the tightly-coupled VI-SLAM following distinct fusion frameworks.

3.2.1 Filter-based

Hesch et al. (2014) derive that the linearization error of EKF will change the unobservable directions of VI-SLAM systems and thus cause inconsistency. The solutions of this issue focus on the linearization point management (Huang et al. 2009), observability constraint (Hesch et al. 2012), the robot-centric formulation (Bloesch et al. 2015; Huai and Huang 2018), and the refinement of the filter (Yang et al. 2022; Van and Mahony 2023). Modifying the EKF linearization point represented by the First-Estimates Jacobian (FEJ) method (Huang et al. 2009) ensures the null space of the observability matrix does not degenerate. OC-EKF (Hesch et al. 2012) explicitly enforces observability constraints. The robot-centric formulation in ROVIO (Bloesch et al. 2015) and R-VIO (Huai and Huang 2018) avoids the alignment with the global gravity vector and the observability mismatch of the world-centric VI-SLAM. In addition, the invariant EKF (Bonnabel 2007) and the equivariant filter (EqF) (van Goor et al. 2023) based on the Lie group have been adopted to circumvent the consistency issue. Van and Mahony (2023) prove that the EqF based on their newly proposed Lie group is a consistent estimator for VI-SLAM with lower linearization error and higher order output. Furthermore, the observability changes when extracted visual features change. Yang and Huang (2018, 2019); Yang et al. (2019a) analyze the uncertainty and unobservable directions of ordinary visual geometric features and their proper expression. They propose two unified forms of visual feature representation: the quaternion and closet point (CP), which is later exploited for CP point and line representations in Yang et al. (2019a), and derive the unobservable subspace for multiple lines and planes (Yang and Huang 2018, 2019).

Generally, filter-based VI-SLAM systems derive from Multi-State Constraint Kalman Filter (MSCKF) (Mourikis and Roumeliotis 2007), following the alternative process of IMU-based propagation and visual-based updates. MSCKF maintains several past cloned camera poses over a sliding window in the state vector instead of feature points and thus avoids the dimensional explosion. A wide range of filter-based VI-SLAM works inherit the framework of MSCKF and enhance it in various aspects (Brossard et al. 2018; Eckenhoff et al. 2019c, a; Heo et al. 2019; Geneva et al. 2019a; Zou et al. 2019; Chen et al. 2023a). Several methods leverage the computation efficiency and high non-linearity from other filters (Geneva et al. 2019a, b; Brossard et al. 2018). Geneva et al. (2019a, 2019b) leverage Schmidt Kalman formulation (SKF) (Schmidt 1966) to treat old keyframes as nuisance parameters and thus reduce the quadratic computational complexity of EKF to $O(n)$. Brossard et al. (2018) exploit the better nonlinear performance of UKF and embed states and errors into a high-dimensional Lie group. Others combine the optimization-based manner with MSCKF for comprehensive use of information. LOMSKF (Heo et al. 2019) adopts optimization in the local window, which contains messages omitted in the original MSCKF such as camera observations. For a thorough understanding of MSCKF and its variants, OpenVINS (Geneva et al. 2020) provides an excellent open platform,⁴ which contains necessary resources ranging from documentation, tools, theories, to projects.

⁴https://github.com/rpng/open_vins.

3.2.2 Optimization-based

The most dominant open-source optimization-based VI-SLAM systems currently are VINS-MONO (Qin et al. 2018) and its variant VINS-FUSION (Qin et al. 2019), ORB-SLAM3 (Campos et al. 2021), and DM-VIO (Stumberg and Cremers 2022). The VINS (Qin et al. 2018, 2019) series are lightweight and easy to equip with embedded devices like Unmanned Aerial Vehicles (UAV). ORB-SLAM3 (Campos et al. 2021) integrates achievements of the ORB series (Mur-Artal et al. 2015; Mur-Artal and Tardos 2017) and supports many modes such as monocular, stereo, their combination with IMU, and RGB-D. However, it relies more on visual features and is prone to fail in challenging environments (Yin et al. 2021). DM-VIO (Stumberg and Cremers 2022) is based on DSO (Engel et al. 2018) and requires photometric calibration like all direct V-SLAM methods.

Commonly used sliding window-based BA (fixed-lag smoother) in optimization-based VI-SLAM linearizes the system by fixing linearization points. This leads to discrepancies with the original optimization and divergence. Besides, its marginalized priors are densely connected with remaining variables, imposing additional computing burdens, which can be amended by selective marginalization (Leutenegger et al. 2015; Qin et al. 2018) and sparsification of dense priors (Hsiung et al. 2018). In addition to improving the sliding-window, several methods increase the computational speed of BA. Liu et al. (2018) offer an incremental BA-based solver and achieves state-of-the-art efficiency at that time. Gopinath et al. (2023) establish GPU-accelerated BA and integrated it with ORB-SLAM3 with local-BA time reduction of 13.81%–33.79%.

3.2.3 Learning-based

Improvements in learning-based VI-SLAM typically exist in the feature extraction and message fusion stage. Some methods select or design feature extraction networks according to the features they prefer. Since CNN is not capable of capturing geometric features critical in state estimation, many learning-based VI-SLAM methods (Cai et al. 2020) have explored networks focusing on geometric features that can promote localization performance. Chen et al. (2019) use FlowNet (Dosovitskiy et al. 2015) for geometrically meaningful image features rather than those related to appearance or context. As for IMU, Liu et al. (2021) tailor a one-dimensional inertial feature encoder for IMU data to avoid the use of time-consuming LSTM.

In terms of fusion methods, the previous VINET (Clark et al. 2017) simply concatenated different features, while current approaches are increasingly moving towards selective fusion methods. For example, Chen et al. (2019) introduces random masking during the feature fusion stage, resulting in weights that can overcome dynamic environmental changes and data interruptions. Besides, they demonstrate that IMU is more robust to fast rotation while visual measurements are preferred in large translations. This is beneficial for determining data ratios in fusion. Liu et al. (2021) exploit attention-guided feature fusion that gets weight from cross-domain attention block and achieves better accuracy than conventional VIN-MONO and ORB-SLAM without loop detection. To raise the generalization of the model, researchers explore the self-supervised or unsupervised learning manners (Shamwell et al. 2018b; Han et al. 2019). Han et al. (2019) take information from the stereo camera as the self-supervision but the feasibility relies heavily on the stereo camera.

3.2.4 Common challenges

As for the general challenges of VI-SLAM, VI-SLAM does not contain depth like LI-SLAM, which makes mapping more challenging. In addition, it contains more explorations of enhancing the utilization of IMU and motion patterns.

Dynamic: Similar to LI-SLAM, visual semantic information (Cui and Ma 2019) plays an important role in the extraction of dynamic features. However, different from DNN filters in LI-SLAM that simply removes potentially dynamic points, VI-SLAM further considers actual dynamics of detected objects (Song et al. 2022a; Samadzadeh and Nickabadi 2023). Song et al. (2022a) adaptively assign weights to features according to reprojection error values to reduce the impact of dynamic outliers. Besides, it proposes a multi-hypotheses-based constraints grouping method to avoid loop detection errors caused by temporal static objects. Samadzadeh and Nickabadi (2023) introduce HRNet (Yuan et al. 2020) as the feature evaluator to determine whether features are dynamic or static. In addition, using IMU measurements is less computationally intensive than using DNNs. Yu et al. (2023) exploit IMU measurements in predicting the position of features and reducing outliers caused by dynamic environments. Zheng et al. (2024) propose an IMU-based dynamic object tracker and combine object detection and bayesian filtering to find static points. Systems with stereo vision can leverage scene flow for dynamic detection (Yin et al. 2023a; Wen et al. 2024a). Yin et al. (2023a) detect dynamic features by comparing IMU-prediction with stereo-generated scene flow. Then they construct virtual landmarks based on the detected dynamics, which are then used to improve the map construction process.

Degeneracy: Similar to LI-SLAM, VI-SLAM systems also detect degeneracy through the quality of visual features (Yin et al. 2024). In addition to degeneracy caused by environments, researchers widely identify degenerative motion through observability analysis (Yang and Huang 2018; Yang et al. 2019a). Since IMU measures the acceleration and angular velocity, it has no output in uniform linear motion and thus leads to divergence of the VI state estimator. Yang and Huang (2018) further derive that VI-SLAM will degenerate when the system moves along the direction of line features or rotates around the gravitational direction.

Online Calibration: Several algorithms attenuate time offset through hardware and only obtain extrinsic online (Zou et al. 2019). Most methods optimize time offset and extrinsic at the same time by involving them in the state vector. Filter-based methods align time-offset-combined IMU propagation and visual pose estimation in the update stage (Eckenhoff et al. 2019a, b; Yang et al. 2019a). Eckenhoff et al. (2019a) interpolate IMU pose into camera frames and estimate calibration parameters by representing the interpolation of unknown time offsets. Optimization-based methods establish factors relative to time and extrinsic (Qin and Shen 2018). Qin and Shen (2018) propose visual feature factors composing time offset based on the constant velocity assumption.

Feature: There are lots of VI-SLAM algorithms that build on feature-based V-SLAM method try to improve feature extraction. Researchers abstract features for specific environments by leveraging their priors (Zou et al. 2019; Qin et al. 2020b; Chen et al. 2023a). For structural places, line features can be attached to the Atlanta world or calculated by vanishing points (Zou et al. 2019). Chen et al. (2023a) propose a planar detection method special for man-made environments. Qin et al. (2020b) deliberately detect semantic features that typically exist in parking lots such as guide signs, parking lines, and speed bumps for

autonomous parking applications. IMU measurements are useful for narrowing down feature extraction and matching (Seok and Lim 2020; Liu et al. 2023). Seok and Lim (2020) use inertial messages to generate soft-pose constraints in the initialization of feature tracking. Liu et al. (2023) combine IMU in optical flow to predict the position of lines.

Not only features specific to particular environments, but visual cues with better quality also matter. Several methods ensure the quality by enhancing the tracking mechanism (Xie et al. 2020) or analyzing motion influences on features (Yu et al. 2023). Xie et al. (2020) propose to seize the effective hierarchical quadtree search mechanism for robust feature tracking. Yu et al. (2023) derive image perspective deformation models to construct an IMU-aided descriptor, which is robust to camera rotation. It is also effective to introduce a feature evaluation module (Zhang et al. 2023c; Jiang et al. 2023). Zhang et al. (2023c) exploit recurrent field transforms from Teed and Deng (2021) to predict dense pixel-wise correspondences between co-visible views and attach confidences to each pixel. Jiang et al. (2023) extract planar features from depth images and corner features from grayscale images. It contains a feature balancing module to identify whether it is a low-texture scene for feature choosing.

Depth Completion: The mapping thread of VI-SLAM systems commonly suffers from the sparsity of features and depth, which drives depth completion a key topic (Wong et al. 2020; Sartipi et al. 2020; Merrill et al. 2021). Deep learning methods are popular in depth completion (Wong et al. 2020; Sartipi et al. 2020; Merrill et al. 2021). Wong et al. (2020) build a two-stage unsupervised dense depth completion network that combines piecewise planar and photometric consistency. Sartipi et al. (2020) complete the depth of the low-texture surfaces from the sparse input and gravity direction. However, it can not meet real-time demand and only focuses on indoor planar environments. Merrill et al. (2021) leverage Fastdepth (Wofk et al. 2019) and achieve real-time state estimation and depth completion. Some researchers exploit information obtained from IMU (Zuo et al. 2021) or the estimated odometry (Usenko et al. 2020). Zuo et al. (2021) combined IMU measurements to provide metric dense depth reconstruction. Usenko et al. (2020) recover non-linear factors from odometry and use them in the global consistent mapping optimization.

Enhance utilization of IMU and motion patterns: Compared to widely studied visual patterns, several recent approaches also take the IMU into account. Buchanan et al. (2023) propose a learning-based IMU estimator that targets IMU bias instead of motion model and is, therefore, able to generalize to unseen locomotion patterns. SR-VIO (Samadzadeh and Nickabadi 2023) adopts a denoising convolutional neural network to eliminate IMU noises and mitigate the divergence issue. The kinematic characters of the system offer additional constraints (Yu et al. 2021; Li and Stueckler 2022). Yu et al. (2021) combine constraints from motion information to confirm stop detection but is only suitable for ground vehicles. Li and Stueckler (2022) include velocity-control-based kinematic motion model (Thrun 2002) constraints and further learns the parameters of it to predict the movement.

3.3 LiDAR-visual SLAM

Since most LV-SLAM systems are designed in a loosely coupled manner and it is inconvenient to categorize them by fusion framework, we divided them into *Visual-centric* and *Modular-type* according to LiDAR utilization.

3.3.1 Visual-centric

Visual-centric LV-SLAM systems normally treat LiDAR merely as the range-augment instrument (Huang et al. 2020; Zhang et al. 2017; Graeter et al. 2018; Shin et al. 2018; Young et al. 2020; Zhu et al. 2021; Li et al. 2021a; Song et al. 2022b; Aydemir et al. 2022). Some methods use dense point clouds to improve depth registration accuracy (Young et al. 2020). DVL-SLAM (Shin et al. 2018; Young et al. 2020) is based on direct V-SLAM to relieve the depth association errors of 3D and 2D points in feature-based one. Extracting key features is also an effective approach to improving LV-SLAM. Huang et al. (2020) introduce point and line features to ameliorate errors caused by noise, large viewpoint changes, and motion blurs.

3.3.2 Modular-type

Modular-type LV-SLAM systems not only combine the distance information but also resort to incorporating the function of LiDAR in SLAM systems by disassembling the systems into modular parts (Zhang and Singh 2015; Seo and Chou 2019; Wang et al. 2021d; Chen et al. 2021a; Chou and Chou 2022; Yin et al. 2022; Yuan et al. 2023; Shu and Luo 2023; Cai et al. 2023). The notable work VLOAM (Zhang and Singh 2015) is composed of a high-frequency LiDAR-assisted depth-enhanced VO and a low-frequency LO, refining the pose from the visual module. Later works tend to adopt the semi-direct V-SLAM methods instead of feature-based ones. SDV-LOAM (Yuan et al. 2023) and LAMV-SLAM (Yin et al. 2022) both based on semi-direct depth-enhanced odometry and pick intermediate frames to achieve better association between 3D and 2D points.

In modular-type LV-SLAM, cameras help LiDAR in many perspectives, such as color completion (Yin et al. 2022; Lin et al. 2024), loop closure (Wang et al. 2021d; Chen et al. 2021a), mapping (Shu and Luo 2023), and assistant constraint (Yin et al. 2022). Lin et al. (2024) colored LiDAR point clouds by projecting RGB pixels to the LiDAR frame. Besides, they maintain a color feature map to constrain pose estimation along with the geometric map. Chen et al. (2021a) detect loop closure by the bag of world model in visual before point clouds verification, which compensates for LiDAR weakness in loop closure detection. Shu and Luo (2023) tailor a multi-modal mapping module that contains LiDAR and camera feature voxels. Yin et al. (2022) comprise an online photometric calibration thread to eliminate photometric disturbances. Based on the integration of the above systems, the general structure of modular-type LV-SLAM is depicted in Fig. 5. Such systems are modular, with LiDAR providing depth information and integrating back-end optimizations from both LiDAR and camera modules.

3.3.3 Common challenges

In terms of common challenges, compared to the previous two types, LV-SLAM is concerned more with the data association process that connects LiDAR and camera measurements.

Dynamic: The dynamic objects can be rejected by alternatively adopting different types of sensors. TVL-SLAM (Chou and Chou 2022) applies a multi-step outlier rejection mechanism. It uses LO as the initial pose to re-project visual map points onto the current image and then rejects matches with large re-projection errors.

Depth-Enhanced V-SLAM

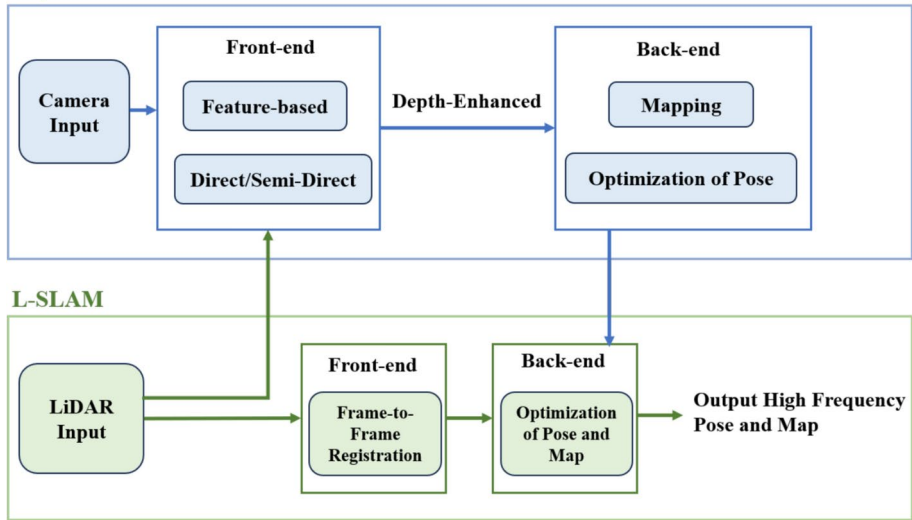


Fig. 5 The general structure of Modular-type LV-SLAM systems

Online Calibration Online calibration of LiDAR and the camera mainly focuses on extrinsic estimation. Some methods achieve this by matching heterogeneous features (Zhu et al. 2021). Zhu et al. (2021) extract the depth, reflectivity, and contour of the LiDAR frame and then compare them with the image contour by ICP.⁵ However, the feature-based calibration requires overlap in FOV, which is not always available. Thus, researchers also leverage the residuals of trajectory to optimize extrinsic parameters (Chou and Chou 2022). For time offset, most LV-SLAM systems achieve synchronization by sophisticated hardware. Tu et al. (2023) mitigate the need for absolute synchronization by splitting pose estimation into two steps. They initiate camera pose by global motion averaging and further joint optimize the separate poses.

Data Association: The data association is commonly implemented on the image plane (Graeter et al. 2018) or an intermediate frame, which is normally a unit sphere centered at the camera (Zhang et al. 2017; Yuan et al. 2023; Yin et al. 2022). LIMO (Graeter et al. 2018) first projects LiDAR points into the image plane to find a rough connections between visual and a set of point clouds. It then performs foreground segmentation to fit local planes and uses it to estimate accurate depth. In DEMO (Zhang et al. 2017), the extracted 2D features are associated with LiDAR depth by projecting them to a unit sphere for data association. Distinct from traditional hand-crafted depth-association methods, learning-based depth completion is also studied (Aydemir et al. 2022; An et al. 2022; Liu et al. 2024). Aydemir et al. (2022) propose a self-supervised hybrid fusion framework to decrease the rotation-only errors. An et al. (2022) convert the point clouds to depth images and then feed them and RGB images into a pose estimation neural network. Liu et al. (2024) propose a network with a bi-directional structure to fuse inconsistent modalities.

⁵<https://github.com/ISSE-Technology/CamVox>.

3.4 LiDAR-IMU-visual SLAM

LIV-SLAM systems are the integration of the aforementioned types and normally contain at least one subsystem of VI-SLAM or LI-SLAM. Early LIV-SLAM systems are mainly dominated by the loosely coupled manner and lack compact connections among diverse modalities (Wang et al. 2019; Khattak et al. 2020; Zhao et al. 2021; Khedekar et al. 2022). Several approaches reinforce the LI-SLAM with the help of VI-SLAM by initializing LiDAR mapping and scan matching using the estimated states of VI-SLAM (Wang et al. 2019; Khattak et al. 2020). VIL-SLAM (Shao et al. 2019) adopts VI-SLAM to initialize LiDAR mapping and leverages LiDAR-based pose estimation to enhance loop closure. It adopts the loosely coupled manner without any cooperation among measurements. More direct approaches parameterize the outputs of VI-SLAM and LI-SLAM as factors under the optimization-based fusion framework (Zhao et al. 2021; Khedekar et al. 2022). SuperOdometry (Zhao et al. 2021) proposes an IMU-centric framework that contains three sub-factor graphs for LI-SLAM, VI-SLAM, and LIV-SLAM. The IMU preintegration factor is used as motion prediction for LI-SLAM and VI-SLAM, and these two subsystems offer constraint factors to bind IMU biases.

Distinct from traditional fusion frameworks, Zhang and Singh (2018) propose a sequential pipeline that processes data in a decreasing frequency mode and formulates joint optimization as multiple small optimizations. The solution exploits a coarse-to-fine manner. The coarse estimation starts with IMU prediction and is then further refined by VI-SLAM and LI-SLAM. Despite the exclusivity of this pipeline, the subsequent tightly-coupled approaches are primarily grounded in filter-based or optimization-based implementations.

3.4.1 Filter-based

Filter-based methods remain similar to the previous process that propagates by IMU and updates with the LiDAR and camera measurements (Yang et al. 2019b; Zuo et al. 2019, 2020; Lin et al. 2021; Lin and Zhang 2022; Zheng et al. 2022; Frosi and Matteucci 2023). Zuo et al. (2019) exploit the framework of MSCKF by preserving cloned past poses of camera and LiDAR. R2LIVE (Lin et al. 2021), R3LIVE (Lin and Zhang 2022), and FAST-LIVO (Zheng et al. 2022) executes IESKF in the update stage. There is no explicit association between LiDAR and camera raw measurements in early LV-SLAM systems. The depth of visual measurements still originates from triangulation in (Zuo et al. 2019, 2020; Lin et al. 2021). Recently, R3LIVE and FAST-LIVO both establish connections between 3D LiDAR map and 2D visual images through photometric error and build a colored point cloud map of the environment. The photometric errors of R3LIVE (Lin and Zhang 2022) are calculated by the residuals of RGB colors. FAST-LIVO (Zheng et al. 2022) is supported by FAST-LIO2 (Xu et al. 2022a) and SVO (Forster et al. 2014) with LiDAR map reuse. It avoids the triangulation and optimization of visual features over multiple keyframes. The specificities of ikdtree structure and the semi-direct method enable it to operate at high speed.

3.4.2 Optimization-based

Optimization-based approaches manage measurements of diverse stages as factors and achieve factor graph optimization (Wisth et al. 2021; Wang et al. 2021c; Jia et al. 2021; Shan

et al. 2021; Tang et al. 2023; Wang and Ma 2023; Wisth et al. 2023; Nam and Gon-Woo 2023; Lang et al. 2023). The factor graph can be specially customized (Wang and Ma 2023) or leverage canonical structures (Shan et al. 2021). mVIL-Fusion (Wang and Ma 2023) proposes a three-level framework that contains a double-sliding-window optimization method. It assigns VI-SLAM as the main window and LiDAR data as the assist window with another window charging for loop closure and global mapping. LVI-SAM (Shan et al. 2021) is the combination of VINS-mono (Qin et al. 2018) and LIO-SAM (Shan et al. 2020) with visual depth augmentation by LiDAR. It improves the robustness of the initialization of VINS-MONO using calculations from LIO-SAM and involves a complete structure of degeneracy detection, loop closure, and data association. Some methods adopt the continuous-time model that enables simultaneous motion distortion removal and trajectory estimation (Lv et al. 2023; Lang et al. 2023). CLIC (Lv et al. 2023) is the first method to utilize continuous-time fixed-lag smoothing with probabilistic marginalization in LIV-SLAM systems. Lang et al. (2023) present non-uniform B-splines where control points are dynamically placed.

3.4.3 Common challenges

As for general challenges, LIV-SLAM incorporates the advantages of the former types and needs to consider the problems contained in each type. **Degeneracy:** Analogous to solutions adopted in LI and CI SLAM, eigenvalues criterion (Shan et al. 2021; Khattak et al. 2020; Lee et al. 2024) and feature quality evaluation (Shan et al. 2021; Wang et al. 2021c) also work in the degeneracy detection of LIV-SLAM. For instance, CompSLAM (Khattak et al. 2020) detects the degenerate direction through eigenvalues of the Hessian of optimization function and changes to the CI submodule when LI degenerates. GR-Fusion (Wang et al. 2021c) exploits the intuitive method that rejects the LiDAR factor when the depth difference between the first and the last frame in the sliding window is less than the threshold. As for visual features, it considers the measurements unstable and ignores the visual factor when tracked features are lower than the threshold. Switch-SLAM (Lee et al. 2024) proposes a non-heuristic method to find the degenerate threshold of the LI submodule by Chi-squared test (Pearson 1992). In addition, researchers migrate knowledge from other domains such as deep learning (Jia et al. 2021; Ju et al. 2022) and fuzzy logic (Nam and Gon-Woo 2023). Livo (Jia et al. 2021) uses the actor-critic method in reinforcement learning to tune sensors' weight adaptively. Ju et al. (2022) propose a learning-based scene-aware error model to evaluate the perception quality and use the result to promote the odometer performance. Nam and Gon-Woo (2023) exploit Type-2 fuzzy logic to build a learnable observation model that handles observation uncertainties and sensor failure issues.

Feature: Researchers consider commonalities between the same feature points captured by LiDAR and camera (Wisth et al. 2021, 2023). Wisth et al. (2021) develop a unified multimodal landmark tracking method for LiDAR and camera measurements, which ensures the uniform representation of 3D line and plane features. In VILENS (Wisth et al. 2023), they extend this representation and integrate it with leg odometry. Zuo et al. (2020) propose a novel planar feature tracking method and divide planar features into SLAM landmark and MSCKF features. They treat those two types with Mahalanobis distance-based outlier rejection in LIC-Fusion2.0.

Data Association: Analogous to LV-SLAM systems, data association that assigns LiDAR depth to corresponding pixels matters in LIV-SLAM systems. The association pat-

terns are also represented by projecting LiDAR point clouds into image planes (Wang et al. 2021c; Wisth et al. 2021) and projecting both measurements in a C-frame (Zhang et al. 2017; Chiodini et al. 2020; Zhang and Singh 2018; Shan et al. 2021; Wang and Ma 2023). A novel voxel-map-based depth-association method was proposed in a vanishing point-aided LIV-SLAM (Wang et al. 2021b). Besides, this work is further integrated into SuperOdometry (Zhao et al. 2021). CoCoLIC (Lang et al. 2023) obtains visual pixel depth from the global LiDAR map by formulating frame-to-map reprojection errors. Hence, it avoids the computationally intensive pixel depth estimation and optimization.

Except for correlating the depth information, handling occlusion or LiDAR stacking is also important in data association. Several methods reject outliers by checking the maximum distance among depth points for a feature (Zhang and Singh 2018; Shan et al. 2021). FAST-LIVO (Zheng et al. 2022) first projects all the points in the visual submap and most recent LiDAR scan to the current frame. Then it examines their depth to check whether there is occlusion. In LE-VINS, Tang et al. (2023) propose a solid state LiDAR-enhanced VI-SLAM which contains a novel depth association method by IMU. They notice that once the depth is obtained, it will be treated as a constant without any refinement or merely a prior in former researches. Hence they avoid outliers through an effective plane-checking algorithm.

In addition, there emerge some efforts to further enhance the correlation between data by exploiting visual measurements (Lin and Zhang 2022; Zheng et al. 2022; Hong et al. 2024). The vectors of tracked points in R3LIVE (Lin and Zhang 2022) contain their 3D position and RGB color. The frame-to-map photometric update relies on the residuals between the predicted and the observed color. After the system converges, the optimized color will be adopted in rendering the texture of the global map. FAST-LIVO (Zheng et al. 2022) attaches patches from the current image to map points within the FOV. LVI-GaussMap (Hong et al. 2024) achieves high-fidelity map reconstruction. It refines the Gaussian LiDAR map by visual photometric gradients.

Online Calibration: Several LIC systems take IMU as the mediator and obtain calibration of IMU with LiDAR and camera respectively (Zuo et al. 2019, 2020). Zuo et al. (2019, 2020) propagate IMU measurements up to the corrected time and involve the extrinsic in the measurement models of LiDAR and camera. Some LIC systems divide calibration into two phases. Yang and Ma Wang and Ma (2022) first calibrate the CI submodule and then linearly interpolate CI-based pose estimation onto the LiDAR for LI calibration. R3LIVE (Lin and Zhang 2022) only calibrates IMU and camera online by containing time offset in the measurement model of the camera. Tong et al. (2024) propose a novel keyframe-oriented data synchronization method to align LiDAR and camera measurements.

3.5 Summary

The summarization of the four types of multi-sensor fusion SLAM is shown in Table 3,4,5,6. In addition, Sect. 3 aims to help readers understand the classic algorithms of the four types of methods and their research differences. Figure 6 is concluded according to the common challenges section of each type of method, intending to summarize the similarities and differences in the research focus of different methods. The following content will expand upon Fig. 6. Through the above analysis, it can be concluded that in addition to refinements of fusion frameworks, multi-sensor fusion SLAM systems commonly concentrate on dynamic processing, degeneracy treatment, online calibration, and feature extraction.

VI-SLAM involves more settlements for depth completion and enhancing the utilization of IMU. LV-SLAM and LIV-SLAM are also concerned with data association between LiDAR and camera measurements.

3.5.1 Dynamic detection and processing

Environment dynamics mainly influence observations of LiDAR and cameras. Both of them can find dynamic points by DNN (Zhao et al. 2019; Samadzadeh and Nickabadi 2023) or geometric constraints (Iio 2021). Containing multi-sensors, dynamic detection can also be achieved by comparing differences of individual measurements (Yin et al. 2023a). The treatment of dynamic has improved in recent years, which ranges from directly filtering out dynamic points (Zhao et al. 2019), exchanging diverse modalities (Seo and Chou 2019; Samadzadeh and Nickabadi 2023), considering temporarily static points (Song et al. 2022a), to maintaining dynamic points as a constraint (Yin et al. 2023a).

3.5.2 Degeneracy detection and processing

Degeneracy can be caused by environments and motions. General fusion frameworks can alleviate it by the covariance matrix of the measurements to some extent. However, this approach will fail when the degeneration is too severe to lead to system breakdown.

Environmental degeneracy influence the performance of LiDAR and camera and is normally detected through eigenvalues of the Hessian matrix of the measurements equation as proposed by Zhang et al. (Zhang et al. 2016). The degenerate directions are related to the eigenvectors corresponding to eigenvalues smaller than the threshold, which is defined through experiments (Khattak et al. 2020; Hinduja et al. 2019; Tagliabue et al. 2021), heuristic methods (Han et al. 2023), and non-heuristic method (Lee et al. 2024). The more intuitive way is checking the quality and number of features (Wang et al. 2021c; Shan et al. 2021; Junior et al. 2022). Some systems also consider message rate (Palieri et al. 2021). In addition, multi-sensor systems can rely on measurements from other sensors. For instance, LIVER leverages VI-SLAM to determine the degenerate directions of LiDAR, while Ground-Fusion (Yin et al. 2024) adopts the wheel odometer for visual degeneracy detection. Modern systems also introduce DNN-based degeneracy detection modules (Jia et al. 2021; Ju et al. 2022; Nam and Gon-Woo 2023). Motion-caused degeneracy has been widely researched in aided-IMU systems by analyzing the null space of the observability matrix (Yang and Huang 2018). As for treatments of degeneracy, some systems simply invalidate data from sensors that are detected to degenerate (Palieri et al. 2021; Tagliabue et al. 2021; Shan et al. 2021; Yin et al. 2024) and constraint the degenerative motions (Yang and Huang 2018; Yang et al. 2022). Systems that detect specific degeneracy directions may optimize the state along with the well-constrained directions (Wen et al. 2024b).

3.5.3 Feature extraction

The extraction of features depends on sensing modalities and the environment prior. There are numerous mature approaches in computer vision, such as corner features (Shi et al. 1994; Leutenegger et al. 2011), edge features (Canny 1986), and line feature detector (Grompone von Gioi et al. 2010). Although researchers have contributed many novel defi-

nitions and descriptions of point, line, and plane features, exploring potential features that are relevant for state estimation is still meaningful. Features existing in a particular scenario can also enhance the performance of the system, such as structural feature extraction in artificial scenes.

3.5.4 Online calibration

Online calibration generally adds extrinsic and time offset parameters into the state vector to optimize together with the pose of the system. For IMU-included systems, original calibration methods minimize the pose estimation discrepancy between IMU propagation and LiDAR or camera odometry (Tagliabue et al. 2021; Le Gentil et al. 2021; Qin and Shen 2018; Eckenhoff et al. 2019a, b; Yang et al. 2019a; Zou et al. 2019; Zuo et al. 2019, 2020; Wang and Ma 2022; Lin and Zhang 2022). Several methods are refined to residuals of angular velocity and acceleration (Zhu et al. 2022a). Most sensor drivers nowadays guarantee a rough clock source synchronization under network protocols once it is plugged into the system. Online calibration allows for more accurate time synchronization. The time offset is always included in measurement functions, such as IMU propagation (Le Gentil et al. 2021) and camera measurement (Qin and Shen 2018; Lin and Zhang 2022) to ensure the time of different sensors is consistent. For LV-SLAM, online calibration mainly focuses on extrinsic estimation through the matching between LiDAR and camera observations (Zhu et al. 2021).

3.5.5 Heterogeneous data association

Data association is mainly the problem in LV-SLAM and LIV-SLAM systems that require the correlation of LiDAR and camera data. Ranging from the simple projection of LiDAR point cloud to image plane (Graeter et al. 2018), optimization based on residuals between 3D map and 2D pixels (Lang et al. 2023), to correlation involving RGB information (Lin and Zhang 2022). The interaction is becoming tighter and more robust.

4 Datasets and evaluate metrics

4.1 Datasets

Open-source datasets avoid complex hardware preparation and deliver a uniform platform to compare the algorithms in diverse scenarios. We collate frequently used datasets in multi-sensor fusion SLAM in Table 7.

Most datasets contain all of the three sensors (Geiger et al. 2012). Some datasets that have been widely used in VI-SLAM evaluation are collected only by cameras and IMU (Burri et al. 2016; Schubert et al. 2018; Wenzel et al. 2021; Minoda et al. 2021). The degeneracy conditions, such as degenerative scenarios and aggressive motion are generally included in current datasets.

Some datasets are oriented toward autonomous driving scenarios and collected by high-speed cars (Geiger et al. 2012; Maddern et al. 2017; Wen et al. 2020; Wenzel et al. 2021). KITTI (Geiger et al. 2012) is a classic dataset collected by a passenger vehicle, involving

Table 3 Characteristics and comparison of LI-SLAM algorithms

References	Fusion framework	Coupled method	Contribution	Open source	Validation environment	Year
LIPS (2018)	Factor	TC	New plane formulation to overcome singularity; opensource LiDAR simulator	✓	Sim and Real	2018
Zhao et al. (2019)	ESKF	TC	CNN dynamic filter; NDT mapping	✗	Sim and Real	2019
Ye et al. (2019)	Factor	TC	Rotation-constrained refinement for scan-to-map alignment	✓	Sim and Real	2019
LINS (Qin et al. 2020a)	IESKF	TC	Robot-centric formulation against filter divergence; ground vehicles	✓	Real	2020
LJO-SAM (Shan et al. 2020)	factor	TC	Local sliding window-based scan-matching to enable real-time performance	✓	Real	2020
LOCUS (Palieri et al. 2021)	–	LC	Conquer sensor failure by LC switching scheme; sensor health monitor of rate-check	✗	Real	2020
LJON (Tagliabue et al. 2021)	Factor	TC	Self-assess its performance and degradation prediction; automatic extrinsic calibration	✗	Real	2020
Fast-ljo (Xu and Zhang 2021)	iEKF	TC	Solid state LiDAR; complexity-reduced Kalman gain; motion compensation	✓	Real	2021
LiLi-OM (Li et al. 2021b)	Factor	TC	Solid LiDAR feature extraction; hierarchical keyframe-based sliding window optimization	✓	Sim and Real	2021
IN2LAAMA (Le Gentil et al. 2021)	Factor	TC	Motion distortion along with state estimation process without an explicit motion model; auto calibration	✗	Sim and Real	2021
MILIOM (Nguyen et al. 2021)	Factor	TC	Use multiple LiDARs to compensate low vertical resolution of a single LiDAR	✗	Sim and Real	2021
CLINS (Lv et al. 2021)	Factor	TC	Non-rigid registration to estimate the continuous-time trajectory; two-stage trajectory correction method	✓	Sim and Real	2021
Li et al. (Li et al. 2021c)	SRIF (Wu et al. 2015)	TC	IMU-aided ground plane extraction; arccos corner points distinguishment	✗	Sim and Real	2021
KFS-LJO (Li et al. 2021d)	Factor	TC	Quantitative evaluation LiDAR features method and feature selection module	✗	-	2021
RF-LJO (ljo 2021)	Factor	TC	Adaptive range image dynamic removal; remove moving objects before state estimation	✗	Sim and Real	2021
FAST-LJO2 (Xu et al. 2022a)	iEKF	TC	Directly register scan and map; ikd-tree for map maintenance	✓	Sim and Real	2022

Table 3 (continued)

References	Fusion framework	Coupled method	Contribution	Open source	Validation environment	Year
Kenji et al. (Koide et al. 2022)	Factor	TC	Voxelized GICP matching cost factor; tightly coupled in global mapping	✗	Sim	2022
EKF-LOAM (Junior et al. 2022)	EKF	TC	Adaptive covariance matrix	✓	Sim and Real	2022
Faster-LIO (Bai et al. 2022)	iEKF	TC	Present iVox to handle point clouds; AA-iEKF; propose iVox-PHC for bunch of points	✓	Sim	2022
Li et al. (Li et al. 2022)	Factor	TC	Feature extraction of solid-LiDAR combine both geometry and intensity information	✗	Sim and Real	2022
LOCUS2.0 (Reinke et al. 2022)	–	LC	A novel normals-based GICP formulation; adaptive voxel grid filter; sliding-window map	✓	Real	2022
Jiang et al. (Jiang and Shen 2022)	iEKF	TC	Uncertainty model and noise estimation for LiDAR measurement	✓	Sim and Real	2022
D-LC-Nets (Liu et al. 2022)	factor	TC	Large-scale point cloud denoising network; learning based loop closure detection	✗	Sim and Real	2022
D-LIO (Chen et al. 2023b)	–	TC	Coarse-to-fine approach to construct continuous-time trajectories for point-wise motion correction	✗	Sim and Real	2023
RI-LIO (Zhang et al. 2023d)	iEKF	TC	Reflectivity image assisted; corrected LiDAR projection model	✗	Real	2023
D-LIOM (Wang et al. 2023c)	factor	TC	Combine gravity information as factors; submap-to-submap loop detection	✓	Sim and Real	2023
DAMS-LIO (Han et al. 2023)	iEKF	LC	Degeneration-aware pipeline; adopt CRLB in verification	✗	Sim and Real	2023
Kim et al. (Kim et al. 2023)	–	TC	Adaptive keyframe generation; kNN based submap generation	✗	Sim and Real	2023
LIO-SEGMOT (Lin et al. 2023)	Factor	TC	Asynchronous state update; hierarchical criterion to couple odometry and object tracking	✗	Sim and Real	2023
SW-LIO (Wang et al. 2023b)	iESKF	TC	Rapid ground segmentation; sliding window based iESKF; novel residuals	✗	Sim and Real	2023
Point-LIO (He et al. 2023)	EKF	TC	High bandwidth; point-by-point state estimation; stochastic process-augmented kinematic model	✓	Real	2023
TransFusionOdom (Sun et al. 2023)	Learn	TC	Leverage advantages of Transformer	✗	Sim	2023

Table 3 (continued)

References	Fusion framework	Coupled method	Contribution	Open source	Validation environment	Year
LIO-GVM (2024)	IESKF	TC	Incremental Gaussian voxel map; variance residual	✓	Sim	2024

Table 4 Characteristics and comparison of VI-SLAM algorithms

References	Fusion framework	Coupled method	Contribution	Open source	Validation environment	Year
ICE-BA (Liu et al. 2018)	Factor	TC	Renovated numerical solver; global consistency	✓	Sim	2018
VINS-MONO (Qin et al. 2018)	Factor	TC	Representative framework covering sliding-window based initialization, optimization and loop detection	✓	Sim and Real	2018
R-VIO (Huai and Huang 2018)	EKF	TC	Sliding-window robocentric VIO; constant unobservable subspace	✗	Sim and Real	2018
Cheng et al. (2018)	EKF	TC	Subspace analysis based temporal stabilization; edge-perserving filters	✗	Sim and Real	2018
Qin et al. (2019)	Factor	TC	Presented a general sliding-window optimization-based framework for multi-sensors	✓	Sim and Real	2019
Geneva et al. (2019a)	EKF	TC	leverage SKF in MSCKF framework to reduce complexity	✗	Real	2019
StructVIO (Zou et al. 2019)	EKF	TC	Atlanta world model; novel line representation integrated Manhattan world	✗	Sim and Real	2019
Yang et al. (2019a)	EKF	TC	CP line representation; sliding window line feature triangulation	✗	Sim and Real	2019
DeepVIO (Han et al. 2019)	Learning	TC	Self-supervised learning-based VIO	✗	Sim	2019
Chen et al. (2019)	Learning	TC	Feature selection based on environment dynamics	✗	Sim	2019
AVP-SLAM (Qin et al. 2020b)	EKF	LC	Semantic feature; semantic map reuse	✗	Real	2020
Vladyslav et al. (Usenko et al. 2020)	Factor	TC	recover non-linear factors from VIO and use in global map optimization	✓	Sim	2020
ORB-SLAM3 (Campos et al. 2021)	Factor	TC	Fast IMU initialization; novel place recognition algorithm; multimap system; abstract camera representation	✓	Sim and Real	2021
VINS-Motion (Yu et al. 2021)	Factor	TC	Consider vehicle motion constrain	✗	Sim	2021
AT-VIO (Liu et al. 2021)	Learning	TC	One-dimension IMU encoder; attention block for fusion	✗	Sim	2021
DM-VIO (Stumberg and Cremers 2022)	Factor	TC	Delayed marginalization and PGBA for IMU initialization; dynamic photometric error weight	✓	Sim	2022

Table 4 (continued)

References	Fusion framework	Coupled method	Contribution	Open source	Validation environment	Year
DynaVINS (Song et al. 2022a)	Factor	TC	IMU-assisted dynamic rejection; multiple hypotheses for temporarily static objects rejection	✓	Sim	2022
KIN-VIO (Li and Stueckler 2022)	Factor	TC	Velocity-control based kinematic motion model; radial basis function kernels compensate for control delay; motion model calibration	✗	Sim and Real	2022
FS-VIO (Yu et al. 2023)	Factor	TC	IMU-aided feature descriptor robust to camera rotation	✗	Sim and Real	2022
BAMF-SLAM (Zhang et al. 2023c)	Factor	TC	semi-pose-graph BA; RPT feature evaluation	✗	Sim	2022
Eq-VIO (Van and Mahony 2023)	EqF	TC	EqF; novel symmetry for VIO; reduce linearization error	✓	Sim	2023
Buchanan et al. (2023)	Factor	TC	Learning-based IMU biases evolution; generalize to unseen motion pattern	✗	Sim and Real	2023
Chen et al. (2023a)	MSCKF	TC	Planar constraint; real-time panar detection	✗	Sim and Real	2023
Gopinath et al. (2023)	Factor	TC	GPU acceleration for BA	✓	Sim and Real	2023
SR-VIO (Samadzadeh and Nickabadi 2023)	Factor	TC	Hybrid visual preprocessing; switchable factors; transformer-based loop clousure	✗	Sim	2023
DDIO-Mapping (Jiang et al. 2023)	Factor	TC	Combien grayscale and depth features; feature filter	✗	Sim	2023
RLD-SLAM (Zheng et al. 2024)	Factor	TC	IMU-based dynamic object tracker; object detection and bayesian filtering for static points	✗	Sim	2024

Table 5 Characteristics and comparison of LV-SLAM algorithms

References	Coupled method	Contribution	Open source	Validation environment	Year
V-LOAM (Zhang and Singh 2015)	LC	High frequency VO; low frequency LO refine	✗	Sim and Real	2015
DEMO (Zhang et al. 2017)	LC	Laser and camera depth association	✗	Sim and Real	2017
LIMO (Graeter et al. 2018)	TC	Local plane for LiDAR-based depth association; semantic outlier rejection	✓	Sim and Real	2018
DVL-SLAM (Shin et al. 2018)	TC	Sliding window tracking; rely sparse measurements; loop closure	✓	Sim and Real	2018
Huang et al. (2020)	TC	Use line features; robust line extraction	✗	Sim	2020
CamVox (Zhu et al. 2021)	TC	LiDAR and camera extrinsic calibration; solid LiDAR	✓	Sim and Real	2021
RPV-SLAM (Kang et al. 2021)	TC	LiDAR for panoramic image range augmentation; panoramic camera	✗	Sim and Real	2021
Self-VLO (Li et al. 2021a)	TC	First learning-based end-to-end Visual LiDAR Odometry; siamese network	✗	Sim	2021
LAMV-SLAM (Yin et al. 2022)	–	Online photometric calibration; two-stage motion estimation	✗	Sim and Real	2021
Song et al. (2022b)	TC	Self-supervised depth completion from LVO; feature map-based Odometry	✓	Sim	2022
TVL-SLAM (Chou and Chou 2022)	TC	New LiDAR residual compression method SLAM-with-calibration; dynamic rejection	✗	Sim	2022
H-VLO (Aydemir et al. 2022)	TC	First hybrid LVO; depth map fusion; densificate sparse LiDAR	✗	Sim	2022
Shu and LUO (Shu and Luo 2023)	TC	Multi-modal feature map with different types of voxels	✗	Sim	2023
SDV-LOAM (Yuan et al. 2023)	TC	New feature extraction and point matching method; semi-direct depth enhance	✓	Sim	2023
PanoVLM (Tu et al. 2023)	–	Panoramic vision and LiDAR fusion no need for strict synchronization	✓	Sim	2023

Table 5 (continued)

References	Coupled method	Contribution	Open source	Validation environment	Year
Lin et al. 2024)	–	RGB colored dense point clouds color and geometric feature constrains	✗	Sim	2024
DVLO (Liu et al. 2024)	TC	End-to-end LV-SLAM local-to-global fusion network	✗	Sim and Real	2024

Table 6 Characteristics and comparison of LVI-SLAM algorithms

References	Fusion framework	Coupled method	Contribution	Open source	Validation environment	Year
Zhang et al. (2018)	Sequential	TC	Unit sphere data association; multilayer processing pipeline	✗	Sim and Real	2018
VIL-SLAM (Shao et al. 2019)	–	LC	LiDAR-aided loop closure; VIO-aided LiDAR mapping	✗	Real	2019
LJC-Fusion (Zuo et al. 2019)	EKF	TC	Online spatial and temporal calibration; LIV in MSCKF framework	✗	Real	2019
Wang et al. (2019)	–	LC	VIO-aided LI-SLAM; multi-modal loop detection; PGO	✗	Sim	2019
CompSLAM (Khattak et al. 2020)	–	LC	VIO prior for LiDAR degeneration	✗	Real	2020
LJC-Fusion2.0 (Zuo et al. 2020)	EKF	TC	Novel plane-feature tracking; observability analysis of LIV; outlier rejection	✗	Sim and Real	2020
David et al. (Wisth et al. 2021)	Factor	TC	Unified representation for LiDAR and visual features; synchronization	✗	real	2021
GR-Fusion (Wang et al. 2021c)	Factor	TC	Dynamic removal; degradation detection	✗	Real	2021
Lvio-Fusion (Jia et al. 2021)	Factor	TC	Segmented optimization; actor-critic self-adaptive method	✓	Sim	2021
SuperOdometry (Zhao et al. 2021)	Factor	LC	IMU centric framework; extensible; dynamic octree; sub-factor-graph	✗	Real	2021
R2LIVE (Lin et al. 2021)	ESIKF	TC	First opensorced LJV-SLAM high rate filter-based odom; low rate optimization-based refines	✓	Sim and Real	2021
LVI-SAM (Shan et al. 2021)	Factor	TC	Failure detection; loop closure; well opensorced structure	✓	Sim and Real	2021
R3LIVE (Lin and Zhang 2022)	ESIKF	TC	LJO for map's geometric structure; VIO render map's texture; 3D map-2D image error; online calibration. RGB contained	✓	Sim and Real	2022
MIMOSA (Khedekar et al. 2022)	Factor	LC	Versatile optimization backend; degeneration detection; prevent out of order updates	✗	Real	2022

Table 6 (continued)

References	Fusion framework	Coupled method	Contribution	Open source	Validation environment	Year
FAST-LIVO (Zheng et al. 2022)	ESKF	TC	Combination of FAST-LIO2 (Xu et al. 2022a) and feature-to-map based SVO Forster et al. 2014); realistic RGB map	✓	Sim and Real	2022
LE-VINS (Tang et al. 2023)	Factor	TC	Robust depth-association method; factor graph for visual features, LiDAR depth and IMU	✗	Real	2023
mVIL-Fusion (Wang and Ma 2023)	Factor	TC	VIO-based LO prediction; double sliding-window; software time sync	✓	Sim and Real	2023
VILENS (Wisth et al. 2023)	Factor	TC	Legged robots; novel factor incorporates joint kinematic measurements; extensive evaluation	✗	Sim and Real	2023
Dinh and Kim (2023)	Factor	TC	AI-assited noise model; Type-2 Fuzzy logic in factor	✗	Real	2023
CLIC (Lv et al. 2023)	factor	TC	Continuous-time fixed lag smoothing multi-LiDAR enable	✓	Sim and Real	2023
Coco-LIC (Lang et al. 2023)	Factor	TC	Non-uniform B-spline; image-to-LiDAR reprojection	✓	Sim and Real	2023
LIV-GaussMap (Hong et al. 2024)	IESKF	TC	High fidelity Gaussian map; compatible with various types of LiDAR	✓	Real	2024
Switch-SLAM (Lee et al. 2024)	Factor	TC	Non-heuristic degeneracy detection, switching-based sensor fusion	✗	Sim and Real	2024

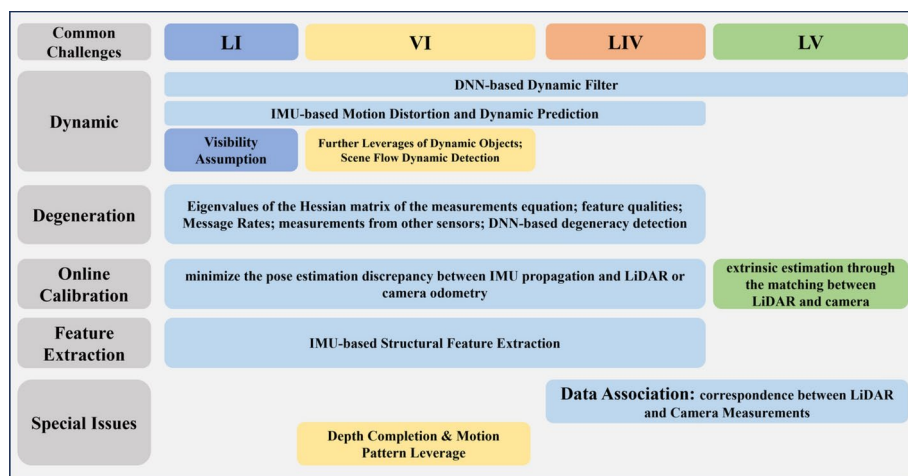


Fig. 6 The conclusion of common challenges in Sect. 3. The light blue boxes show how different methods address the same issues. Gray boxes indicate common problems. Other colored boxes highlight issues specific to each type

urban scenes and textureless highways. It is worth mentioning that the synchronized low frequency IMU data in KITTI is not suitable for testing multi-sensor systems with IMU. Oxford RobotCar dataset (Maddern et al. 2017) contains 1-year driving data, involving long-term weather changes and building variations. UrbanLoco (ulhk) (Wen et al. 2020) is collected in highly urbanization areas, containing urban canyons, bridges, and tunnels. 4Seasons (Wenzel et al. 2021) contains 350 km of recordings in nine environments, ranging from tunnels to highways. Some datasets are collected by a relatively slow Ground Robot (GR) (Carlevaris-Bianco et al. 2016; Yin et al. 2021; Lacroix et al. 2020). North Campus Long-Term (NCLT) dataset (Carlevaris-Bianco et al. 2016) is collected on a campus covering 147.4 km over 15 months. Erfoud (Lacroix et al. 2020) is designed for planetary exploration tasks and was collected by two ground robots in three distinct Mars-like terrains. It consists of nine different trajectories recorded at various times of the day. M2DGR (Yin et al. 2021) contains challenging situations not included in previous datasets, such as entering lifts and complete darkness. Beyond that, it involves sequences with back-and-forth and zig-zag movements. Datasets collected via UAV (Burri et al. 2016; Nguyen et al. 2022) or handheld devices (Schubert et al. 2018; Ramezani et al. 2020; Wei et al. 2024) deliver more motion patterns. NTU-VIRAL (Nguyen et al. 2022) is obtained by UAV on campus, containing trajectories such as UAV movement as it turns and rises. Newer College (Ramezani et al. 2020) involves data of highly dynamic motions in textureless open space. Fusion-PortableV2 (Wei et al. 2024) contains diverse platforms ranging from a handheld device, a quadruped robot (QR), and a vehicle. Hilti-Oxford dataset (Zhang et al. 2023b) is collected by handheld device in indoor environments, containing floor variations and limited texture.

There are varying levels of dynamics that exist in datasets. Autopilot scenarios naturally include dynamic objects. However, most of them contain only medium levels of dynamic disturbances, with occasional pedestrians or vehicles passing by (Geiger et al. 2012; Maddern et al. 2017). Some datasets are less dynamic, with only some sequences containing slow-moving objects (Yin et al. 2021; Zhang et al. 2023b). VOIDE (Minoda et al. 2021) is

Table 7 Datasets for multi-sensor fusion SLAM

Datasets	Sensor			Environment		Collected	Dynamics	Degeneracy
	LiDAR	IMU	Camera	Other	Indoor	Outdoor		
KITTI (2012)	HDL-64E	✓	✓	GNSS	–	Urban	Car	Highway; high speed
Oxford Robotcar (2017)	HDL-64E	✓	✓	–	–	Urban	Car	Long-term weather changes; building variations
NCLT (2016)	HDL-32E; UTM-30LX	✓	✓	GNSS	Campus	Campus	GR	High changing light levels; snow
EuROC (2016)	–	✓	✓	–	Machine Hall; Vicon room	–	UAV	Sharp turns; bobbing up and down
TUM-VI (2018)	–	✓	✓	MoCap; Light sensor	Campus	Campus	Hand	Corridors, in-out changes
Erfoud (2020)	HDL-32E; HDL-32	✓	✓	GNSS	–	Planetary	GR	Bumpy ground, planetary geology, textureless large scenes
4Seasons (2021)	–	✓	✓	GNSS	Parking lot	Urban; Highway; Rural	Car	Challenging weathers; tunnels and highways
Urbanloco (2020)	HDL-32E	✓	✓	GNSS	–	Urban	Car	Very tall buildings; urban canyons, bridges, and tunnels; driving on cross-harbour road
Newer College (2020)	Ouster-64	✓	✓	–	–	Campus	Hand	Textureless open space; aggressive angular motions
UrbmRobotcar (2020)	HDL-32E	✓	✓	GNSS; Radar	–	Urban	Car	Snow, sloping road, right overtaking
VIODE (2021)	–	✓	✓	–	Parking lot	City	UAV	Extensive FOV obstruction
M2DGR (2021)	VLP-32	✓	✓	Infrared camera; Event camera; GNSS	Campus	Campus	GR	Back-and-forth; zig-zag; lift; textureless hall
NTU-VIRAL (2022)	Ouster-16	✓	✓	UWB	Campus	Campus	UAV	6DOF drastic movements;
Hilti-Oxford (2023b)	Hesai-32	✓	✓	–	Buildings	–	Hand	Dark corners, long corridors, narrow stairs

Table 7 (continued)

Datasets	Sensor LiDAR	Environment			Collected	Dynamics	Degeneracy
		IMU	Camera	Other			
FusionPortableV2 (2024)	Ouster-128	✓	✓	Event camera	Campus	Indoor	Outdoor
					Campus	Campus	Campus
MCD (2024)	OS1-64	✓	✓	UWB	Hand; QR; Car	Hand	high
	Livox Mid-70				Hand	Low	grassland, escalator, underground tunnel glass reflection, extreme lighting

captured in simulation and primarily used to evaluate methods incorporated with dynamic processing. UtbmRobocar dataset (Yan et al. 2020) contains cyclical variation and highly dynamic objects. Recent datasets have started to involve measurements of solid-state LiDAR. For instance, Multi-Campus Dataset (MCD) (Nguyen et al. 2024) contains scans of Livox Mid-70.

4.2 Evaluation metrics

The evaluation metrics of a SLAM system focus on its accuracy and efficiency. The efficiency is evaluated by the runtime of the algorithm. For accuracy, the Absolute Trajectory Error (ATE) and the Relative Pose Error (RPE) are commonly used (Schubert et al. 2018), which are usually calculated by EVO⁶ tool. ATE measures the global deviation between the estimated trajectory and the true trajectory, which is useful to test the global consistency. RPE quantifies the local accuracy of the trajectory within a fixed time interval Δ , such as revealing local drift per meter. They are both calculated by the root mean square error (RMSE) and can be shown as:

$$\begin{aligned}
 ATE_{RMSE} &= \sqrt{\frac{1}{N} \sum_{i=1}^N \|trans(T_{gt,i}^{-1} * T_{est,i})\|^2} \\
 RPE_{RMSE} &= \sqrt{\frac{1}{N-\Delta} \sum_{i=1}^{N-\Delta} \|trans(T_{gt,i,\Delta}^{-1} * T_{est,i,\Delta})\|^2} \\
 T_{gt,i,\Delta} &= T_{gt,i}^{-1} * T_{gt,i+\Delta} \\
 T_{est,i,\Delta} &= T_{est,i}^{-1} * T_{est,i+\Delta}
 \end{aligned} \tag{13}$$

where $T_{gt,i}$ and $T_{est,i}$ represent the real and estimated pose at i -th timestamp.

For very long sequences the evaluation is typically conducted by segmenting the trajectory into fixed-length intervals, with the metrics calculated for each segment. Take the evaluation methods of KITTI (Geiger et al. 2012) for example. $t_{rel}(\%)$ (The average translational RMSE drift (%) on lengths of 100 m-800 m) and $r_{rel}(\%)$ (the average rotational RMSE drift ($^{\circ}$ /100 m) on lengths of 100 m-800 m) are commonly used metrics, which can be calculated as:

$$\begin{aligned}
 t_{rel}(\%) &= \frac{1}{N} \sum_{i=1}^N \frac{\|t_i - t_{gt,i}\|}{\|t_{gt,i}\|} \times 100 \\
 r_{rel}(\%) &= \frac{1}{N} \sum_{i=1}^N \frac{\angle(R_i^T R_{gt,i})}{d} \times 100
 \end{aligned} \tag{14}$$

where N is the total number of segments in the trajectory, t_i and R_i represents the translation and rotation estimation at i -th time step, respectively. $t_{gt,i}$ and $R_{gt,i}$ represents the corresponding ground truth of translation and rotation. d is the segment length.

⁶<https://github.com/MichaelGrupp/evo>.

The mapping performance is still qualitatively judged by manual observation. With the emergence of methods with 3D Gaussian and NERF, Peak Signal to Noise Ratio (PSNR) and Structural Similarity Index (SSIM) are also adopted in mapping evaluation (Hong et al. 2024).

5 Future challenges

The following section proposes several future challenges to formulate a more versatile multi-sensor fusion SLAM system.

5.1 Make full use of deep neural networks

DNNs have the potential to extract the implicit connection between a diversity of data, which is revealed in other fields (Chitta et al. 2022). It inspires us to explore their possible capacity in multi-sensor fusion SLAM systems.

5.1.1 Combination of geometric and semantic feature

Traditional methods merely extract artificial features represented by points, lines, and planes. These features are not stable in unstructured or resolution-changing scenes (Lianos et al. 2018) and are deficient in environmental comprehension. However, DNNs can grasp both geometric and semantic features according to specialized loss functions. For geometric clues, DNNs generally capture the optical flow in the image stream (Teed and Deng 2021; Zhang et al. 2023e), which still lacks the understanding of the scenarios. Semantic clues exploited in semantic SLAM systems can facilitate dynamic filtering and loop detection but are commonly parallel with the geometric thread, omitting its contribution to localization. Thus, simultaneously extracting geometric and semantic features by a network enables achieving an end-to-end multi-sensor fusion SLAM system with scene interpretation. Moreover, DNNs have been productive in multimodal data processing in other domains, such as object detection and semantic segmentation (Chitta et al. 2022; Zhang et al. 2023a). The output of these DNNs and the ideas they employ can be migrated into multi-sensor fusion SLAM. Nonetheless, the generalization of learning-based feature extraction is affected by datasets. The more datasets DNNs are trained on, the more scenarios they can adapt to. However, the actual computing power is limited. The Large Model⁷ may address this issue to some extent.

5.1.2 Pose regression

Pose regression means regressing the pose by DNNs, which is first practiced by PoseNet (Kendall et al. 2015). Researchers improve it by adding LSTM layers (Walch et al. 2017) or exploiting Transformer (Shavit et al. 2021) in pose inference. Sattler et al. (2019) reveal that pose regression is more related to image retrieval and desires refinements for better accuracy. Data from various sensors may contribute to enriching the statistical support for pose regression and obtaining better results.

⁷<https://chat.openai.com/>

5.2 Enhance and exploit map information

The present research prefers strengthening the pose estimation and often neglects the enhancement of the mapping function with multi-sensors.

5.2.1 Map quality

Traditional map representations are discrete and can be categorized as depth point, height, surfel, and volumetric. With the development of implicit mapping and neural radiance fields (NeRF) (Mildenhall et al. 2021), several V-SLAM algorithms employ this continuous representation to render the scenarios (Zhu et al. 2022b; Zhang et al. 2023e). GO-SLAM (Zhang et al. 2023e) enables real-time implicit neural mapping in large-scale indoor environments and achieves more precise localization results on some datasets where ORB-SLAM3 fails. Recent works also combine 3D Gaussian splatting (Hong et al. 2024). As mentioned in 2.6, high-fidelity map representations such as NeRF and 3DGS have begun to trickle into single-sensor SLAM systems (Zhu et al. 2024b; Matsuki et al. 2024).

5.2.2 Map application

Since most systems solely leverage the maps in global optimization during a single run, the subsequent value of maps obtained by multi-sensor SLAM systems is seldom mentioned. These maps contain ample metric or even semantic details and can be branched out to multiple downstream missions, ranging from navigation and post-disaster exploration to augmented reality human-computer interaction. Besides, the multi-sensor configuration can support a more comprehensive perception of the world. Arm et al. (2023) exploits the multi-sensor fusion SLAM algorithm in the scout robot which is in charge of exploring the planetary analog environment. The robot offers the environmental map for other robots on planetary exploration, demonstrating the promising scope of expanding applications of maps reconstructed via SLAM.

5.3 Fuse more modalities and analyze optimal combinations

Although the integration of LiDAR, IMU, and camera is the most common configuration in present multi-sensor fusion SLAM systems, combinations with other modalities can provide information unavailable in current systems. The extensible modalities are pretty diverse, ranging from GNSS (Cao et al. 2022), Ultra Wide Band (Nguyen et al. 2022), to event cameras (Zhou et al. 2021). Especially with the widespread implication of quadrupedal robots and wheeled vehicles, the combination with wheel encoders (Hua et al. 2023; Guadagnino et al. 2024) and leg odometry (Yang et al. 2023) has become increasingly popular. Apart from that, invoking sensors from neighboring infrastructures also has the potential to promote multi-sensor fusion SLAM systems. For instance, in places supported by wireless communication signals, robots performing exploration tasks can access information from surveillance cameras in the buildings as an auxiliary to the SLAM system.

However, while it is de-facto that multi-sensor fusion SLAM systems deliver better performance than those equipped with fewer sensors, the noise and computational burden that accompany multiple modalities should not be overlooked. We believe that there exist opti-

mal combinations of modalities, both in terms of the number and types of them. Existing theoretical analyses primarily concentrate on the consistency and invariance of the fusion framework or dependent features. They yield the optimal solution in a specific manner rather than investigating the necessity of a supplementary sensor. Hence, to prevent the abuse of sensors and maximize the effectiveness of each sensor, it is imperative to analyze whether there is an upper threshold to the benefits afforded by additional modalities and where the threshold lies.

5.4 Design novel general fusion framework

A critical subject in multi-sensor fusion SLAM systems is determining how to integrate a wide range of sensor measurements, namely the selection of a fusion framework. The conventional fusion frameworks based on filtering or BA have dominated and spawned a series of accurate and efficient algorithms with the learning-based fusion progressively coming to prominence. However, these types of fusion frameworks prioritize purely perceptual effects and neglect the requirements of the missions as well as the limitations of the scenarios. Hence, designing a task-driven fusion framework that adaptively configures and selects the sensors suitable for the demands of scenarios and assignments is essential.

5.5 Combination of another application

The extent application of SLAM can cooperate with object tracking that contains a mutual facilitating effect, which is called SLAMMOT (Feng et al. 2023; Lin et al. 2023; Bahraini et al. 2018, 2019). By denoting the dynamic obstacles in bounding boxes, the system leverages the dynamic elements to assist self-localization rather than directly discarding them. Meanwhile, the SLAM results will correspondingly boost the performance of object tracking, empowering the system to fulfill more intricate tasks, i.e. decision and planning in autonomous driving.

Henning et al. (2023) further broadens this incorporation to binding with human motion tracking and performs better than ORB-SLAM3 in datasets containing people. Not only vehicles and humans but virtually all kinetic objects in the environment can be exploited, and it is an open problem of developing a generic paradigm for modeling them and what trade-offs ought to be imposed for various dynamic objects. Air-ground coordination and cross-view matching (Shi et al. 2019) are also considerable extensions since they can offer extra localization and environmental information.

In addition, multi-sensor fusion SLAM algorithm could also combine with the application of Multi-Robot and Cooperative systems. Multi-robot collaborative SLAM systems are often employed in large-scale exploration tasks, such as planetary exploration (Sotodeh Bahraini et al. 2021; Bahraini et al. 2021). In such systems, each robot is responsible for exploring a specific region of the environment, and the sub-maps are eventually merged to form a comprehensive global exploration result (Fang and Li 2024). Multi-source fusion SLAM algorithms can be applied to each robot individually, thereby improving the overall exploration quality.

Moreover, multi-sensor fusion SLAM can integrate semantic recognition information with geometric data for practical applications. Kimera (Rosinol et al. 2020), through its modular structure, combines fast geometry-based localization with semantic 3D map recon-

struction, successfully enabling the creation of interpretable maps in unknown environments. DSP-SLAM (Wang et al. 2021a), on the other hand, effectively utilizes semantic priors from the map to aid in localization and prior map construction. These applications demonstrate the powerful potential of semantic information within the multi-sensor fusion SLAM framework.

5.6 Reduction of computational consumption

SLAM algorithms are often supported as localization modules for unmanned systems. However, the systems may also contain modules for decision-making, battling, and planning. These modules share the arithmetic power of the on-board computer, leading to the problem of resource hogging. Thus it is necessary to design multi-source fusion SLAM algorithms with high computational efficiency and low energy consumption. It provides stable localization and map-building support for unmanned systems with minimal consumption.

5.7 Combination with large language model

Integrating Large Language Models (LLMs) into multi-sensor fusion SLAM offers a promising avenue for future research. LLMs have advanced capabilities for contextual understanding, reasoning, and decision-making. These can enhance SLAM systems by facilitating semantic understanding of unstructured environments, improving data correlation between heterogeneous sensors, and enabling advanced task planning in dynamic scenarios. There are already some works that combine LLM with SLAM. For example, LP-SLAM (Zhang et al. 2024) utilizes LLM to extract language information in the environment to assist positioning. OrionNav (Devarakonda et al. 2024) leverages LLMs to enable intelligent path planning based on SLAM. Thus, LLMs' ability to learn and generalize from large datasets could enable SLAM systems to handle new environments with minimal retraining, paving the way for smarter and more autonomous agents. Exploring these possibilities could significantly advance the field of multi-sensor fusion SLAM.

6 Conclusion

This paper is intended to provide an overview of research in multi-sensor fusion SLAM systems in recent years and present a comprehensive summary of the components and underlying requirements. The multi-sensor fusion SLAM systems are composed of odometry and mapping and the contributions of multi-sensor configuration are ubiquitous in odometry. The research works in recent years have centered on tightly coupled systems and fusion frameworks have been dominated by filter-based and optimization-based approaches with the emergence of several learning-based fusion algorithms. Disparities of sensors lead to differences in feature extraction, residual representation, and the necessity of spatial and temporal calibration, which can be completed offline and online.

We categorize the algorithms according to the fused sensor types and identify that there are commonalities in their improvements, which primarily lie in refinements of fusion frameworks, feature extraction, dynamic processing, degradation detection, online calibration, and data association. Furthermore, we collate the public datasets frequently used for perfor-

mance evaluation and propose several future challenges that a high-level long-term stable multi-sensor fusion SLAM system may encounter. We hope this paper will foster additional exploration into combining the complementary strengths of heterogeneous sensors.

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